A Note on Foundations of Statistics[1]

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Contents

Chapter 1

Savage's *Foundations of Statistics*

1.1 Sketch of Savage's Theory

Savage's theory is formulated by 7 axioms, known as P1-P7. P1 asserts that there is a binary relation $\preceq$ the agent holds, which is a weak order[1] to be interpreted as "not preferred to", over acts. This order is the one and only primitive notion Savage explicitly assumes. The existence of such an order, however, assumes, for one, the existence of acts of a certain kind, which in turn assumes the existence of other things. All remaining 6 axioms serve the purpose of putting further constrains on the structure of this weak order, so that Savage can get the desired resulting theory, namely the now standard calculation of expected utility, via representation theorems.

Acts, for Savage, are functions from potential states to consequences. The space of states $S = \{s_1, s_2, ..., s_n, ...\}$ and the set of all consequences $\{C_1, C_2, ...\}$ are, hence, also primitive for Savage. An event E is a subset of S. In general, I will use capital letter E possibly with subscripts, $E_1, E_2, ...$ for events. Any (total) function from the state space to the set of consequences, is an act. I will use lower case letters $f, g, ...$ for acts. $f \preceq g$ thus means "f is not preferred to g". Savage requires and indeed only needs two consequences to which the agent is not indifferent. Denote them by B and W, for "better" and "worse" respectively.

To compare the agent's personal probabilities on an arbitrary pair of events, E_1 and E_2, construct two acts: f_{E_1} and g_{E_2}. f_{E_1} returns B if E_1 obtains and

[1] A weak order, as I would use the term in this note, is an order that is transitive, complete and (hence) reflexive. Savage calls this kind of order 'simple order'. Contemporary terminology seems to recommend calling it 'total preorder' or 'non-strict weak order'.

returns W if $\neg E_1$ obtains; similarly, g_{E_2} returns B if E_2 obtains and returns W if $\neg E_2$ obtains. We can induce the agent's personal qualitative probability over these two events, based on whether the agents prefers f_{E_1} and g_{E_2} or neither. After obtaining qualitative probability in this way, based on P6, Savage used certain standard mathematical tricks provided by Measurement Theory (See [Krantz et al., 1971]) to induce quantitative probability from qualitative probability. The utility function is defined from the same set of primitives by piggybacking on [Von Neumann and Morgenstern, 1947]. That is, Savage proved that all axioms of von Neumann and Morgenstern follow from his P1-P7, and hence utility functions can be defined in the exactly same way as it's done by von Neumann and Morgenstern.

1.2 Terminology

Savage uses the term 'partition' as a technical term to mean what we now call 'finite partition' (p.24). It's an unfortunate choice since Savage later proposes P7 to address acts defined on infinite partitions (known as non-simple acts, in contrast to simple acts, which are acts defined on finite partitions of the state space). As a consequence of this unfortunate usage of the term, he was forced to avoid the term 'partition' in his discussion of P7 (pp. 76 - 78). It's important to keep in mind this particular usage of the term 'partition', since considering infinite partitions throughout will raise problems not addressed in this book of Savage: (1) when infinite partitions are allows, Savage's theory runs into the problem of conditional probability; namely, there will be too many null-events and Savage's P2 doesn't allow one to conditional on null-events (see Section 1.4.3); (2) to provide a general theory of probability, Savage, like de Finetti, assumes only finite additivity but not countable additivity; when the event space is allowed to be infinite, finitely additive probability faces the problem of non-conglomerability (see Section 2.4).

In this note, I will use $\preceq$ for the primitive binary relation among acts (and hence among consequences, with consequences taken as constant acts); P for quantitative probability; $\mathcal{P}$ for qualitative probability; $\sqsubseteq$ and $\sqsubset$ for comparing qualitative probability, $\leq$, $<$, $>$ and $\geq$ for comparing quantitative probabilities and the ordinary order relation on numbers.

1.3 P1

P1 assumes that there is a weak order[2], which is transitive, complete and (hence) reflexive, to be interpreted as non-strict preferences ("not preferable over") over all actions (i.e. all functions from the state spaces to outcomes).

[2]Savage calls this kind of order 'simple order'.

P1 and P2 are the two postulates of Savage doing most of the work. P1 is intuitively not very plausible for requiring agents to have preference over all acts. Denying P1 typically makes certain acts incomparable to each other, in contrast to Savage theory, where any two acts are comparable. One way to deny P1 is given by Issac Levi in his theory of imprecise probability, discussed later in Section 6.3.5.

1.3.1 Real-valued Representation

A weak order alone doesn't suffice for real-valued representation. (See [Krantz et al., 1971] for more.) A famous counterexample is as follows:

Take the set $\mathbb{R} \times \{0, 1\}$. Define order R over it by $(x, y)R(x', y')$ iff $x \leq x'$ or $x = x' \wedge y = 0$.

Suppose h is a map from $\mathbb{R} \times \{0, 1\}$ to $\mathbb{R}$ that preserves the order R, i.e. $(x, y)R(x', y')$ if and only if $h((x, y)) \leq h((x', y'))$.

Consider $(x, 0)$ and $(x, 1)$ for some $x \in \mathbb{R}$. By definition, $(x, 0)R(x, 1)$ but not $(x, 1)R(x, 0)$. Then, $h((x, 0))$ must be strictly less than $h((x, 1))$. Then the closed interval $[h((x, 0)), h((x, 1))]$ is non-empty. For any $x \neq y$, $[h((x, 0)), h((x, 1))]$ and $[h((y, 0)), h((y, 1))]$ are disjoint. Hence $\{[h((x, 0)), h((x, 1))] | x \in \mathbb{R}\}$ is a partition of $\mathbb{R}$.

Such partition, however, isn't possible. Given for any $x \in \mathbb{R}$, by elementary real analysis, there is a rational number $\hat{x}$ in the interval $[h((x, 0)) < h((x, 1))]$. This construction hence gives us uncountably many distinct rational numbers, which is contradictory.

To get a real-valued representation, in addition to a weak order, one needs an Archimedean condition. For Savage, this Archimedean condition, together with further conditions, is delivered by P6.

1.3.2 Sen's properties α, β and γ

Let S be a menu, which is a set of options (finite or infinite). S^+ be a superset of S, i.e. a menu extended from S. $\mathcal{C}$ is a choice function that takes a menu as input and gives a subset of that menu (to be interpreted as the selected/preferred ones) as output. For convenience, we often omit the brackets {,} when applying $\mathcal{C}$ to certain sets of elements. For example, instead of $\mathcal{C}(\{x, y\})$, which is type-wise correct, we write $\mathcal{C}(x, y)$ for convenience. Notice that C doesn't assign any structure to the selected items in the menu. All selected ones are equally preferred.[3]

We can define a binary relation R from a choice function $\mathcal{C}$ as follows: yRx iff for some menu S,[4] $x \in \mathcal{C}(S)$ and $y \in S$. That is, yRx iff there is a menu

[3]For more on choice functions, see Sen's influential paper [Sen, 1971].

[4]Typically, the quantifier ranges over all finite menus over all options. The result doesn't

where x is chosen in the presence of y (and y may or may not be chosen). Under the assumption that $\mathcal{C}$ satisfies property α and β, R is a weak order equivalent to Savage's $\preceq$.

Def. Property α (a.k.a. Chernoff condition): For any menu S, $x \in S, x \notin \mathcal{C}(S) \Rightarrow x \notin \mathcal{C}(S^+)$.

Property α says that you cannot promote an inadmissible (unchosen) option in the menu to be admissible by adding items to your menus.

Def. Property β: For any menu, $(\forall S)x, y \in \mathcal{C}(A)$, $A \subset B$, $y \in \mathcal{C}(B) \Rightarrow x \in \mathcal{C}(B)$.

Property β says that if two things are both selected in some menu A, then in a larger menu B containing A, they are either both selected or both not selected.

α and β together are equivalent to Savage's P1, in the sense that if the choice function $\mathcal{C}$ satisfies both α and β, then the ordering R induced from $\mathcal{C}$ is a reflexive, transitive, complete ordering. It can be seen from these two different presentations of P1 that P1 requires the agent to have a transitive, complete, context-free preference over acts.

We may also induce a choice function $\hat{\mathcal{C}}$ from the induced ordering R by defining $\hat{\mathcal{C}}(S)$ as the set $\{x|x \in S \text{ and } \forall y \in S, yRx\}$ for every S. Without certain restrictions on $\mathcal{C}$, $\mathcal{C}$ and $\hat{\mathcal{C}}$ can fail to be identical.

For example, suppose there are three options x, y, z and $\mathcal{C}(x, y) = x$, $\mathcal{C}(x, z) = x$, $\mathcal{C}(y, z) = y$ and $\mathcal{C}(x, y, z) = y$. By definition of R, we known that $R = \{(x, y), (y, x), (x, x), (y, y), (z, z), (z, x), (z, y)\}$. Then, $\hat{\mathcal{C}}(x, y) = \{x, y\} \neq \mathcal{C}(x, y)$.

We call a choice function $\mathcal{C}$ *normal*, if $\mathcal{C}(S) = \hat{\mathcal{C}}(S)$ for all menu S.

Notice that $\hat{\mathcal{C}}$ is induced from the ordering R by pair-wise comparison. An option is admissible in a menu iff it doesn't lose to any other option in the menu in pairwise comparison. A choice function being normal is equivalent to its being essentially binary in composition, that is the choice function can be recovered from pairwise comparisons.

Sen shows that satisfying property α and β is sufficient but not necessary for the choice function to be normal. Being normal is equivalent to a weaker set of properties, α and γ.

Def. Property γ: Let M be a set of menus, and let V be the union of all sets in M. Then if for all $S \in M$, $x \in \mathcal{C}(S)$, then $x \in V$.

Property γ says that if a choice is admissible in every menu of a class of menus, then that choice must be admissible in the union of all those menus. γ property allows us to determine "the winner of grand tournament" by "local tournaments", and in particular by pairwise (or other local) comparison.

Property γ is weaker than β in the presence of α in the sense that $\alpha + \beta \vdash \gamma$

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change if we consider infinite menus as well. It's also possible to restrict the set of all menu in certain ways. But it's irrelevant for our purpose. For more on this see Section 6 of [Sen, 1971].

and $\alpha + \gamma \nvdash \beta$. Like β, γ is also adding certain context-freeness to the agent's preference. α and γ are taken to be important by economists since these two conditions together allow us to generate preference without worrying about the context of comparisons. Much later in Section 6, we will see that most economists decide to reject P2 instead of P1 when departing from Savage's theory, in order to keep α and γ.

1.3.3 An example against property α

Seidenfeld raises the following example against the α:

There is a restaurant Teddy often goes to. There are two main dishes available: meat and fish. Teddy overall prefers fish to meat. However, the restaurant has two chefs, the good one and the bad one. Only one of the chef is in charge of the kitchen at a time. But the bad chef is in charge the most of time and always overcooks the fish, which makes it less preferable than the meat option. Teddy wants to order fish when the good chef is in charge. But when the bad chef is in charge, Teddy would prefer meat instead. However, Teddy cannot know directly which chef is in charge. He finds it impolite to ask or go into the kitchen. As a consequence, Teddy always orders fish as the main dish. One day, the waiter presents Teddy an extended menu, containing a fancy desert that isn't not often available. On that day, Teddy orders the fish and is happy with his choice, because from the availability of the fancy dessert, he knows that the good chef is in charge.

In this example, Teddy violates the α, but apparently isn't acting irrationally. One may object that in this case, the choices (fish or meat) are only weakly but not strictly connected to what Teddy will get. If we finer-grain the choices to {good fish, bad fish, meat}, then the α won't be violated. This is exactly the point of Seidenfeld's objection. There is uncertainty intra-choice in additional to uncertainty inter-choice. Since the choice on the menu never comes with perfect detail, whenever we make a choice, we face intra-choice uncertainties, that are not characterized by preference relation among choices. Sometimes, menu can provide us information about the intra-choice uncertainty. As a result, α is violated.

Savage's theory cannot characterize such uncertainty, and, a fortiori, cannot model the process how an agent learns information about such uncertainty from the menu.

1.4 P2

P2 intuitively says that when comparing two acts, the cases where they are identical, no matter (so to speak) to which constant act they are identical to in those cases, do not matter at all.

P2 is another weight-lifting postulate of Savage. It gives Savage act-state independence as well as dominant principle (reasoning by cases). Act-state independence is a pre-requisite of dominance principle (comparison by cases). However, we know that there are cases where the dominance principle and, hence, the act-state independence fails. In a two players game, where player 1 acting in certain ways can bring about player 2 acting in certain way, the dominance principle doesn't always lead to optimal results. Situations of this kind cannot be modeled in Savage's theory. One way to solve this problem is by taking conditional preference (in contrast to unconditional ones, which Savage takes as primitive).

When departing from Savage's theory, most economists choose to reject P2 and typically arrive in Prospect Theory discussed later in Section 6.4, in order to keep P1 or at least part of P1, namely α and γ.

1.4.1 Allais Paradox

Allais paradox was originally proposed against von Neumann and Morgenstern. However, this problem is also a problem for Savage, since P1+P2 doesn't allow one to make Allais choice. The assumption most blamed for this failure has been P2 in the past. (See Savage [Savage, 1954], pp. 101-102.)

Allais' paradox relies on the existence of three distinct payoffs, to be interpreted as poor, satisfyingly rich and extremely rich. In Savage's formulation, the three consequences are chosen as \$0, \$500,000, and \$2,500,000 respectively. Due to inflation, one may have to adjust the numbers to make it work.

Savage reconstruct Allais' paradox in terms of lottery tickets. There are 100 tickets and with equal probability, one draws 1 ticket from 100 tickets.

	1	$2-11$	$12-100$
Gamble 1	5	5	5
Gamble 2	0	25	5
Gamble 3	5	5	0
Gamble 4	0	25	0

Table 1.1: Savage's reconstruction of Allais's paradox

Define 4 gambles as in the above table.

Allais argues that when one has to choose from the first two gambles, one would prefer Gamble 1 over Gamble 2, because he won't want to take the risk of not being rich, for the chance of being extremely rich. When one has to choose from the second two gambles, one would prefer Gamble 4 over Gamble 3, since Gamble 4 intuitively is almost as likely as Gamble 3 but makes one extremely rich instead of merely satisfyingly rich. We may call this pair of choices "the

Allais choice". However, one cannot make the Allais choice without violating P2 (supposing P1).

P2, intuitively says that cases where two acts bring about identical payoffs (agreeing cases) do not matter for the decision, not matter what the payoff they bring in agreeing cases. Both the first two gambles and the second two gambles agree on cases 12-100. When making comparison, if one prefers Gamble 1 over Gamble 2, one has to prefer Gamble 3 over Gamble 4, and vice verse.

Savage's solution to this problem is to reject the intuition Allais is relying on, and argue P1 and P2 helps him correct an error he made in his initial reaction. According to his "personal taste", he would prefer Gamble 1 over Gamble 2, and hence Gamble 3 over Gamble 4.

1.4.2 Ellsberg Paradox

Ellsberg paradox [Ellsberg, 1961] was devised specifically against Savage's theory. (The same Ellsberg who released the Pentagon Papers.)

Suppose that there are in total 90 chips in an urn, among which, 30 are red and the remaining is either blue or white. We construct four gambles as in the following table.

	Red	Blue	White	Utility
Gamble 1	1	0	0	1/3
Gamble 2	0	1	0	0-2/3
Gamble 3	1	0	1	1/3 - 1
Gamble 4	0	1	1	2/3

Table 1.2: Ellsberg paradox

Like in the Allais paradox, per P1 and P2, the case of White should be ignored and if one prefers 1 over 2, one has to prefer 3 over four. The intuition Ellsberg relies on is that it's seemingly reasonable for one to prefer 1 over 2 and to prefer 4 over 3.

The takeaway message from the Ellsberg paradox is that personal probability alone, unlike what Savage claims (that personal probability exhausts all uncertainty one faces when making decisions) isn't enough to characterize all uncertainties one faces when making decisions. Additional notions, like risk, should be introduced.

Both Allais paradox and Ellsberg paradox both rest on fairly robust intuitions. It might be worthwhile to note a debunking response against Ellsberg, due to Jay Kadane. Kadane thinks that people make the kind of decisions advocated by Ellsberg because they don't really believe the setting. They (tend to) think that if someone is offering you such a game in such a format to you, they must be trying to trick you in one way or another. Had people really believe in the setting, they will not make the Ellsberg choices and hence will not violate P2.

1.4.3 Conditional Probability

Conditional probability is an important issue which Savage didn't discuss too much in his book. One reason is that Savage's book is meant for a broad audience beyond mathematicians, while issues of conditional probability get technical rather quickly. Also, the result that every finite additive probability can be extended so that probabilities are defined on the whole space, including conditional on null-events, wasn't proved until [Krauss, 1968] and [Dubins et al., 1975].

Judging from his discussion of P2, Savage seems to be aware of those subtleties involved in conditional probability. His pre-theoretical discussion of P2 clearly involves conditional comparisons and conditional preferences. However, he decides to use unconditional preferences to simulate such conditional considerations.

In Kolmogorov probability theory, conditional probability can only be taken on sub-σ-algebra but not on events. Thus, the conditional probability on a particular event (typically null events, i.e. events with measure zero) can depend on the choice of sub-σ-algebra, that is, depend on how you parametrize the state space. Kolmogorov is well aware of this issue and uses the Borel-Kolmogorov paradox to illustrate this 'feature' of his probability theory.

De Finetti is among the first ones to disagree with Kolmogorov on this issue. De Finetti's mainly disagree with Kolmogorov on two points: (i) de Finetti prefers to assume finite additivity only while Kolmogorov assumes countable additivity; (ii) de Finetti proposes that probability should be conditional on events rather than sub-σ-algebra.

De Finetti stipulates the following condition for finite additive conditional probability[5]:

For any finite additive probability P and any event E, F, G such that $EG \neq \emptyset$, $FG \neq \emptyset$:

(i) $P(EF|G) = P(E|F,G)P(F|G) = P(F|E,G)P(E|G)$;

(ii) $P(\cdot|G)$ is a finite additive probability and $P(G|G) = 1$.

However, there is no guaranty that for any conditional probability P and any non-empty event G, such conditional probability $P(\cdot|G)$ exists until [Krauss, 1968] and [Dubins et al., 1975] showed these conditions are necessary and sufficient for a finite additive probability to be generated to a full conditional probability.

Conditional probability is a problem for Savage as Savage only takes uncon-

[5] These conditions are occasionally attributed to Karl Popper for his early discussion of axioms of probability in 1938.

ditional preference over acts as primitive. Conditional probability in Savage's theory (or indeed everything in Savage's theory) must be reduced to unconditional preference. Such reduction is done using called-off acts following de Finetti's idea on called-off gambles.

I will illustrate this method using the following example.

	E_1	E_2	E_3	E_4
A_1	c_{11}	c_{12}	c_{13}	c_{14}
A_2	c_{21}	c_{22}	c_{23}	c_{24}
A_3	c_{31}	c_{32}	c_{33}	c_{34}

E_1, E_2, E_3, E_4 form a partition of the state space and A_1, A_2, A_3 are three acts as defined in the table above. P1 imposes a primitive (unconditional) preference $\preceq$ over A_1, A_2, A_3 and all other acts. From this unconditional preference, the agent's personal unconditional probability over events can be derived, as in demonstrated in Section 1.1.

If we want to get conditional probability, say conditional on event E_1, we make all acts uniform on the complement of E_1, in this case, $E_2 \cup E_3 \cup E_4$. This process transforms all acts A we previously have into A^*, which is identical to A on E_1 and have some arbitrarily chosen value c on the complement of E_1, as shown in the following table. P2 guarantees that the choice of c doesn't make a difference.

	E_1	E_2	E_3	E_4
A_1^*	c_{11}	c	c	c
A_2^*	c_{21}	c	c	c
A_3^*	c_{31}	c	c	c

Notice our primitive (unconditional) preference is also defined on these stared acts (A_1^*, A_2^*, A_3^*, etc.). Indeed, it's defined on all acts. We then can defined a conditional preference relation $\preceq_{E_1}$ s.t. for any act A and B, $A \preceq_{E_1} B$ iff $A^* \preceq B^*$. So long as this relation $\preceq_{E_1}$ is a weak order, it will generate a probability as $\preceq$ does. This probability $P_{E_1}(\cdot)$ generated from the underlying unconditional probability P elicited from $\preceq$ and the event E_1 is just the conditional probability $P(\cdot|E_1)$ of P on E_1.

So far, this method of reduction seems okay. However, it prevents us from conditional on null events. On page 24, Savage notes that "[An event] B is **null** if and only if, for all [acts] f and g, f $\preceq$ g given B". That is, an event is null if and only if the conditional preference relation it generates is degenerate in the sense that no act is preferred over another. Such degenerate ordering, although nevertheless is still a weak order, fails to satisfy Savage's P5, that there are at least two consequences (which are constant acts) that the agent are not indifferent from. Any null event, hence, fails to qualitative, a fortiori, quantitative probabilities.

One may find the definition Savage gives for null events a bit surprising, as this definition directly puts Savage into trouble. The intuitive meaning of a null event is simply an event of measure zero. Take the real line as the state space. Consider the finite partition of this space consists of event $Z(ero)$, event $P(ositive)$ and event $N(egative)$, where Z contains only one real number 0 only while P and N contain all positive and negative real numbers respectively. The event Z is clearly a null but non-empty event. Why cannot my primitive preference relation distinguish acts that only differ on the point 0? Say act A returns 1000 on Z and 0 everywhere else and act B returns 10 on Z and 0 everywhere else. Why cannot my $\preceq$ distinguishes A from B? If so, conditional preference generated by Z, a null event, wouldn't be degenerate.

In fact, such difference is too fine to be represented by real-numbers. One must endorse some kind of dominance principle for this difference to be drawn, which Savage explicitly denies in page 39, so that his theory can yield real representations, i.e. real-valued expectations, for every act.

This problem of conditional on null events will arise even if only finite partitions of the state space are allowed. The example above is a clear illustration of this. Of course, if your underlying state space is finite, an event is null iff it's empty and the problem of null events will not arise. However, as we will see later in Section 1.6, Savage's P6 forces the underlying state space to be infinite and forces every state to be null.

De Finetti's usage of called-off gambles to simulate conditional probability faces exactly the same problem as Savage does here. It's hard to see how this problem of conditional probability can be solved without adding new primitive, e.g. conditional preference in addition tofull-blown preference.

1.5 P3-P5

P3-P5 are assumed by Savage to obtain qualitative probability, in the way sketched in the very beginning of this chapter. For an arbitrary pair of events, E_1 and E_2, construct two acts: f_{E_1} and g_{E_2}. f_{E_1} returns B (better) in case of E_1 and returns W (worse) in case of $\neg E_1$; similarly, g_{E_2} returns B in case of E_2 and returns W in casee of $\neg E_2$. Then define that $\mathcal{P}(E_1) \sqsubseteq \mathcal{P}(E_2)$ iff $f_{E_1} \preceq g_{E_2}$.

P3-P5 intend to make consequences state independent by putting constrains on them, so that a unique qualitative probability can be generated from an agent's preferences. However, as we will see in this section, P3-P5 together didn't succeed the task. When modeling the same agent (i.e. the same set of preferences), different qualitative probabilities can be generated when different (state dependent) consequences are chosen. It should be emphasized that this flaw of Savage's theory, as it seems to us, is not a mere oversight of Savage. In fact, Savage's theory doesn't have enough machinery to overcome this flaw.

The uniqueness of qualitative probability and the uniqueness of quantitative probability cannot be achieved within Savage's framework, using unconditional preferences as the only primitive. New primitives are needed to solve this problem.

P5 is an auxiliary axiom requiring that there are at least two consequences the agent is not indifferent with. In this following, we will analyze P3 and P4 in turn.

1.5.1 P3

P3: If $f \equiv C_1$ and $f' \equiv C_2$, and E is not null; then $f \preceq f'$ given E, if and only if $C_1 \preceq C_2$.

$f \equiv C_1$ is defined as f being a constant act that returns C_1 in every event E. Consequences (C_1 and C_2) are compared by comparing corresponding constant acts.

Intuitively, P3 says that if one prefers C_1 over C_2 in some non-empty event E, then in any event, one prefers C_1 over C_2. By P3, Savage intends to achieve the independence between acts and consequences.

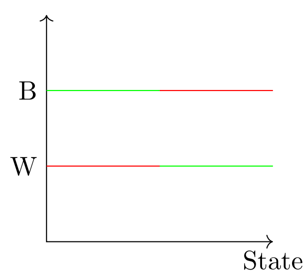


Figure 1.1: Reversal consequences are ruled out by P3

Seidenfeld points out that P3 only requires non-reversal of consequences in different cases, but doesn't force consequences to be "constant". Consider Savage's example of bathing suits and tennis racks. Savage suggests that bathing suits and tennis racks are not suitable as consequences under the uncertainty that one may go to the beach or the tennis club for vacation. Intuitively, bathing suits and tennis racks together violate P3. One will prefer bathing suits if he is going to the beach for vacation, while he will prefer tennis racks if he's going to the tennis club for vacation.

However, suppose one is very enthusiastic about tennis that he prefers tennis racks over bathing suits regardless where he's going for vacation. However, he may still prefer tennis racks more in the case that he's going to the tennis club than he does in the case that he's going to the beach. That is his preference over bathing suits and tennis racks never reserve, but his preference of tennis racks does vary over cases.

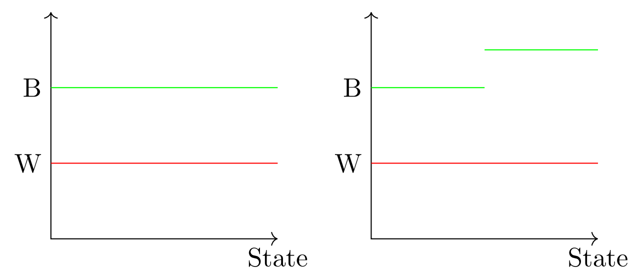


Figure 1.2: Consequences that satisfy Savage's P3

P3 was intended to force the set of outcomes to be constant in the way shown in the picture on the left of Figure 3. P3 successfully rules out consequences, as shown in Figure 4, that reverse in different situations. But doesn't exclude non-constant non-reversal consequences shown in the right of Figure 3.

Judging from Savage's discussion in Section 5.5 of his book, Savage might be vaguely, but certainly not clearly, aware of the problem. To address this problem, he in addition proposes P4 to rule out more unwanted choices of consequences. Unfortunately, P4 doesn't rule out all unwanted choices either. From our current point of view, the job (of establishing state-consequence independence) cannot be done using unconditional preferences as the sole primitive.

1.5.2 P4

P4 requires that the qualitative probabilities generated by any two pairs of consequences are identical. It successfully rules out a certain portion of, but not all, unwanted choices of consequence.

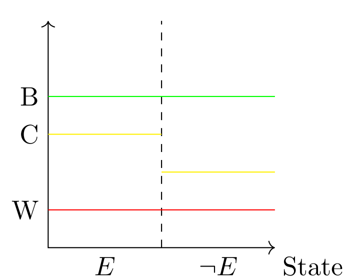


Figure 1.3: This choice of consequences is ruled out by P4

Here, I illustrate How the set of consequences as displayed in Figure 5 is ruled out by P4. Suppose the consequence C, represented by the yellow line in Figure 5, varies in the case of event E and its complement $\neg E$, that C is more preferable in the case of $\neg E$ than in E. (Such comparison, of course, cannot be made within Savage's framework. To criticize Savage's theory, we have to step out of Savage's framework and assume certain things as, so to speak, what is behind the curtain. By reflecting on what Savage's theory tells us in such stipulated cases, we can understand what advantages and disadvantages Savage's theory has.) E and $\neg E$ together form a partition of the state space.

Take an agent, who, as a matter of fact, takes E and $\neg E$ as equally probable. If we construct the probability over E and $\neg E$ using B and W as outcomes (via f_E and $f_{\neg E}$), we will get that E and $\neg E$ are equally probable for the agent, since the agent won't have a strict preference over this two acts. If we construct the probability over E and $\neg E$ using C and W as outcomes (via g_E and $g_{\neg E}$), we will get that E is more probable than $\neg E$, for g_E would be strictly preferred over $g_{\neg E}$. These two different choices of consequences, therefore, give different qualitative probability. Namely, when using B and W, E and $\neg E$ are judged to be equally probable; when using C and W, E is judged to be (strictly) more probable.

In this case, when choosing B, C, W as displayed in Figure 5 as consequences, for this agent, who has a consistent unchanging belief, that E and $\neg E$ are equally probable, constructing personal probabilities using different consequences yields different personal probability. This choice of consequences, thus, is ruled out from Savage's theory for violating P4.

However, P4 does not rule out all "unwanted" choices. For example, when there are only two consequences, P4 is automatically satisfied. Also, when consequences fluctuate in a harmonic manner, P4 is also satisfied.

All four choices of consequences shown in Figure 1.4, satisfy P1-P4, as well as P5, which merely requires that there are at least two consequences the agent is not indifferent with. Indeed, any two consequences, not matter how their relative values fluctuate across states, so long that they never reverse, will satisfy P1-P5.

This subtle failure of state-consequence independence leads to devastating problems for Savage. Although unintended, Savage's theory nonetheless admits state-dependent consequences. This raises devastating problems for Savage's theory, as different choices of consequences whose relative values fluctuate in different ways will generate different qualitative probability and hence different quantitative probabilities for the very same agent.

1.5.3 State Dependent Utility

Suppose the agent, as a matter of fact, takes E and $\neg E$ as equally probable. Using the pair of consequences on the top-right corner of Figure 1.4, we will get a model of Savage's theory that suggests that $\neg E$ is more probable than E

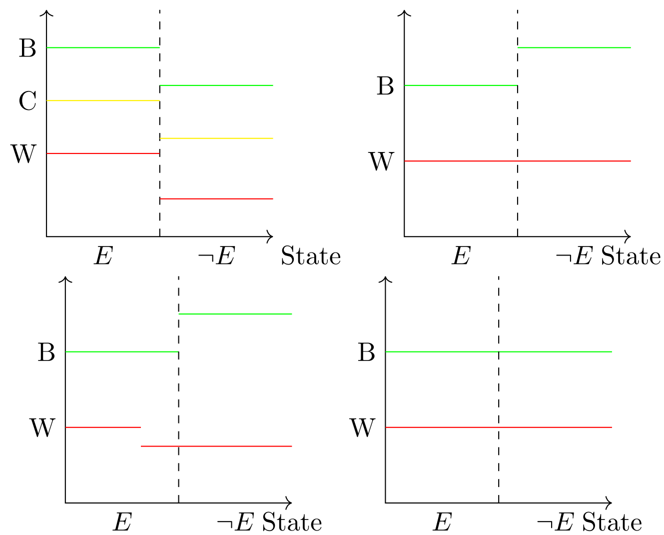


Figure 1.4: Some choices of consequences that satisfy P1-P5

for the agent, while using the pair on the bottom-right corner of Figure 1.4, we will get a model of Savage's theory that suggests that E and $\neg E$ are equally probable for the agent.

Thus, two models of Savage's theory for the same agent yield different qualitative probability, and therefore, will also yield different quantitative probability.

This is **the problem of decomposition**. The intended model of Savage's theory is apparently the second one, where all consequences are state independent and the model correctly elicits the agent's judgment that E and $\neg E$ are equally probable. In the first model, which although unintended, is nevertheless a model of Savage's theory, the agent prefers to bet on $\neg E$ (i.e. prefers the act $f_{\neg E}$ over f_E), not because she judges $\neg E$ to be more probable than E, but because she prefers B at $\neg E$ over B at E. This model of Savage's theory falsely decomposes the agent's preference of $f_{\neg E}$ over f_E into the agent's probability judgment, by assuming all consequences are state-independent.

This objection to Savage's theory may be called an intra-model objection, where the first model under discussion is apparently unintended. Savage's theory simply fails to rule it out. This objection also comes at an inter-model level. As we can imagine, most realistic consequences are state-dependent in the non-reversal sense: their relative value often fluctuate if different states of the world obtain. Choosing different consequences will then yield different qualitative and hence quantitative probabilities for the same agent.

This objection can be made clearer by considering qualitative probabilities instead (more details on qualitative probability will be given in later sections).

As one might expect, Savage's main representation theorem simply is that $f \preceq g$ iff $\sum_i P(s_i)U(f(s_i)) \leq \sum_i P(s_i)U(g(s_i))$, where U is a state-independent utility function and $f(s_i)$, $g(s_i)$ are the constant act corresponding to the outcomes of f and g respectively at state s_i.

Since Savage fails to rule out non-reversal state-dependent consequences, Savage will be forced to have state-dependent utilities. That is the same consequence indeed gives the agent different utilities at different states.

Once state-dependent consequence utility is allowed, the utility function U can give the same consequence different utility at different state. Thus, we can also index U by i and obtain $f \preceq g$ iff $\sum_i P(s_i)U_i(f(s_i)) \leq \sum_i P(s_i)U_i(g(s_i))$, where the value of the utility function U_i depends on the state s_i where the consequences obtain.

For any probability Q that is mutually absolutely continuous with P (i.e. $Q(E) = 0$ iff $P(E) = 0$, for every E), the following also holds (almost trivially): $f \preceq g$ iff $\sum_i Q(s_i d)V_i(f(s_i)) \leq \sum_i Q(s_i)V_i(g(s_i))$, where $V_i(\cdot) = \frac{P(s_i)}{Q(s_i)}U_i(\cdot)$. That is for any two mutually absolutely continuous probabilities P and Q, we can construct state dependent utilities functions U and V such that (P, U) and (Q, V) are both representations of the same set of preference.

Thus we almost completely lose control of the personal probability generated by a given preference relation. So long as we choose consequences carefully enough, any two mutually absolutely continuous probabilities can represent the same preference relation.

One may argue among all choices of consequences allowed by Savage, some choice is apparently better than some others. But it's not clear that there is, so to speak, "the true choice of consequences" that will generate "the correct personal probability". This makes choosing the set of consequences for Savage's theory similar to choosing "rigid rods" in physics. In physics, choosing different rigid rods will give us different geometries. Here, choosing different consequences will give us different probabilities.

A (not so realistic) real-world concern one may raise from this result is as follows: Suppose The World Bank wants to improve global financial stability by requiring major banks in the world to hold an amount of liquidity in proportion to the risk of their investment. But the measurement of risks will be currency dependent. (The jargon is numeraire.) The risk of an investment will turn out to be different using different numeraire (basic standard by which value is computed). Suppose the bank made an investment in rice production. Suppose most rice is produced and consumed in China. The same investment will be rated to be less risky if CNY is chosen as numeraire, instead of USD. For if evaluated by USD, the fluctuation in currency exchange rate will become part of the risk of the investment.

Both intra-model and inter-model decomposition problems cannot be solved within Savage's framework. With unconditional preference as primitive only, it's hard to see how non-reversal state-dependent consequences can be avoided. The apparently solutions to this problem are either to fix P or to fix U by taking more primitive notions.

[Karni et al., 1983] takes conditional preferences as primitive in addition to unconditional preferences. Once we have conditional preferences, we will be able to pin down U, for we can distinguish the choice in the top-right conner of Figure 6 from the choice in the bottom-right conner of Figure 6 by asking the agent's preferences over the constant act B conditional on E and on $\neg E$.

[DeGroot, 2005], on the other hand, pins down P by taking the notion "one event being at least as likely as another" as primitive and provides axioms implying the existence of a unique subjective probability distribution over states of nature.

Small World

In Section 5.5, Savage runs into the problem of "small worlds", where he considers whether he can apply his theory to decisions in limited settings, e.g. whether to take umbrella, without taking all preferences (or equivalently all decisions to be made) into consideration.

Savage ends up thinking that personal probabilities generated by his theory in a small world will depend on what algebraic structure one assigns to that small world, which is apparently devastating for his theory.

[Schervish et al., 1990] thinks that the problem Savage runs into is simply the problem of state-dependent utility as presented above in 1.5.3. The diagnosis Savage gives is simply mis-guided by all the technical details he gets lost in.

1.6 P6

P1-P5 gives us qualitative probabilities from the primitive binary relation $\preceq$. P6 makes the move to obtain quantitative probabilities from qualitative ones.

In fact, P1-P6 gives quantitative probabilities for simple acts, i.e. acts that only have finitely many different outcomes. The main result Savage achieves with P1-P6 is the following: for any two simple acts f and g, $f \sqsubseteq g$ iff $\sum_{j=1}^{k} P(E_j) * U[f(E_j)] \leq \sum_{j=1}^{k} P(E_j) * U[g(E_j)]$.

Recall that we use $\sqsubset$ and $\sqsubseteq$ for comparing qualitative probability $\mathcal{P}$ and use $<$ and $\leq$ for comparing quantitative probability P.

We say that there is a **strict agreement** between qualitative probability $\mathcal{P}$ and quantitative probability P iff for any two events E and F, $P(E) \leq P(F) \Leftrightarrow \mathcal{P}(E) \sqsubseteq \mathcal{P}(F)$; and there is a **almost agreement** between $\mathcal{P}$ and P iff for any two events E and F, $P(E) < P(F) \Rightarrow \mathcal{P}(E) \sqsubset \mathcal{P}(F)$. Intuitively, in the case of almost agreement, the quantitative probability P is faithful for the qualitative probability $\mathcal{P}$, which has a richer structure than P.

For finite space (i.e. only finitely many states), P1-P5 don't even suffice for almost agreement between qualitative and quantitative probabilities. A counterexample, originally due to [Kraft et al., 1959], is reconstructed in [Fishburn, 1986].

Savage avoids this problem (the solution to which wasn't known to him) by forcing the state space to be infinite and imposes an Archimedean axiom to for real-valued representation, using P6.

Savage's move can be understood in the following way. We can get quantitative measurements from qualitative measurements, with arbitrary concatenation. The main different between qualitative measurements and quantitative measurements is that for quantitative measurements, we have concatenation.

Axioms for Extensive Measurement ($\oplus$ is a concatenation operator):

A_1: (Weak Order) $\preccurlyeq$ is a weak order (i.e. transitive and connected).

A_2: (Weak associativity) $O_1 \preccurlyeq O_2$ iff $O_1 \oplus O_3 \preccurlyeq O_2 \oplus O_3$.

A_3: (Monotonicity) $(O_1 \oplus O_2) \oplus O_3 \approx O_2 \oplus (O_1 \oplus O_3)$ (where $\approx$ means both $\preccurlyeq$ and $\succcurlyeq$).

A_4: (Archimedian) If $O_1 \prec O_2$, then for any O_3, O_4, there exists a positive integer n such that $nO_1 \oplus O_3 \preccurlyeq nO_2 \oplus O_4$, where nO_1 is defined inductively as: $1O_1 = O_1$, $(n+1)O_1 = nO_1 \oplus O_1$.

A_5: (Positivity) $O_1 \preccurlyeq O_1 \oplus O_2$ for any O_1, O_2.

$A_1 - A_5$ is necessary and sufficient for (i) $L(O_i) \leq L(O_j)$ iff $O_i \preccurlyeq O_j$ and (ii) $L(O_i \oplus O_j) = L(O_i) + L(O_j)$.

With extensive measurement (A_1 to A_5), we can measure things, in terms of our unit, up to arbitrary accuracy using unbounded concatenations. For example, if we only to measure O_2 in terms of O_1, we can pile m copies of O_1 and n copies O_2 up, until these concatenated copies are of exactly the same length (by qualitative comparison), then $L(O_2)$ is just $(m/n)L(O_1)$.

However, this option of unbounded concatenation isn't available to Savage, as probability is clearly bounded by 1. To get around of this problem, Savage took an alternative approach by requiring that things can be broken into parts as wish. Intuitively, if we can always evenly break things (say O_1) into two pieces, we will be able to measure things (say O_2) in terms of it up to arbitrary accuracy.

P6 says that for any consequence C, for any pair of acts f and g such that $f \prec g$, there is a finite partition $\{X_i | 1 \leq i \leq k\}$ of the state space, such that for any f^* that is exactly f except for some j $(1 \leq j \leq k)$, $f^*(X_j) = C$, and for any g^* that is exactly g except for some j' $(1 \leq j' \leq k)$, $g^*(X_(j')) = C$, $f^* \prec g$ and $f \prec g^*$.

P6 serves multiple purposes for Savage.

(1) P6 forces the state space to be infinite, as C can be arbitrarily large and f and g can be arbitrarily close two each other without being identical. In order for P6 to hold, the space must be arbitrarily divided, which makes it infinite. As a consequence, no finite state space will satisfy P6.

(2) P6 also forces every state in the state space to be null, which raises the problem of conditional probability discussed in Section 1.4.3. Suppose there is state w^*, which is not null. We use numbers as consequences and assume the higher the better. Then take acts A and B that gives (say) 1 and 2 respectively at w^* and 0 everywhere else. Then, intuitively, $A \prec B$. By P6, picking any consequence c, say 1000, there is a finite partition of the state space, such that changing the return of either A or B in any element of that partition to c doesn't change the preference relation between A and B. However, w^* must be in one of the elements of that partition. Given w^* is not null, making A equal to c on that element w^* is in will make A preferable over B. Therefore, every state in the state space must be null.

(3) P6 precludes both infinitely large and infinitely small consequences, (but allows arbitrarily large finite consequences). If C as a consequence has infinite (large or small) value, given two acts $f \prec g$ that only obtain finite value at each state, the P6 will again be violated.

1.7 Utilities

The main result Savage aims at is the quantitative representation theorem that $f \preceq g$ iff $\exists$ unique[6] (P, U) $\int_{s \in S} P(s)U(f(s))d_pP \leq \int_{s \in S} P(s)U(g(s))d_pP$, where U is state-independent[7] utility function.

Savage obtains this result by piggybacking on von Neumann and Morgenstern, proving every axiom von Neumann and Morgenstern assumed is a theorem in his theory. Before giving a survey for von Neumann and Morgenstern's result, I discuss several apparent differences between Savage and v-M and how Savage resolves these differences in his theory.

It's easy to notice that the preference relation in Savage has a finer structure than it in von Neumann and Morgenstern. For example, suppose the agent judges E and E^c to be equally probable. Take f and g such that $f(E) := c_1$, $f(E^c) := c_2$ and $g(E) := c_2$, $g(E^c) = c_1$. For von Neumann and Morgenstern, f and g are the same gamble, while for Savage, they are two different acts. For this reason, Savage on page 70, shows that all acts identical as v-M gambles are equally preferred. He then embraces the concept of (v-M) gambles as equivalent classes of acts and induces a weak order over gambles from the weak order over acts he takes as primitive.

Another potential issue with v-M that may concern Savage is that, as we

[7] $f(s)$ and $g(s)$ are constant acts corresponding to the outcomes of f and g on s. Therefore, contrary to its appearance U is supposed not to depend on s.

will see later, von Neumann and Morgenstern take gambles to be closed under convex combination. Savage has to make sure that his act space (the set of all acts) is large enough to capture all gambles von Neumann and Morgenstern make use of. To solve this problem, Savage stimulates that all functions from states to consequences are acts. This stipulation certainly gives Savage (perhaps more than) enough acts to accommodate all of v-M gambles. But it also leads to the small world concern, to be discussed in Section 1.5.3.

von Neumann and Morgenstern take more primitives than Savage. In addition to the preference relation $\preccurlyeq$ on gambles (or lotteries, in v-M terminology), they also take probability distribution as primitive to define gambles: a simple gamble L is a probability distribution over a finite set of outcomes $\{c_1, ..., c_n\}$. Note a gamble as a sequence $\langle p_1, p_2, ..., p_n\rangle$, where $p_j \geq 0$ for all j and $\sum_{j=1}^{n} p_j = 1$. The goal of von Neumann and Morgenstern is to get the quantitative representation of preferences over finite outcomes (i.e. utility function $U : \{c_1, ...c_n\} \to \mathbb{R}$) such that:

$$L_1 \preccurlyeq L_2 \text{ iff } \sum_{j=1}^{n} p_{1j}U_j \leq \sum_{j=1}^{n} p_{2j}U_j \tag{*}$$

where $L_1 = \langle p_{11}, ..., p_{1n}\rangle, L_2 = \langle p_{21}, ..., p_{2n}$ are two arbitrary gambles.

(*) is the *expected utility property.* A function U is called an expected utility function if it satisfies (*). Utility functions are unique up to positive linear transformation. Take any function U' that satisfies (*). There is a pair $\alpha, \beta \in \mathbb{R}$, such that $U' = \alpha U + \beta$.

von Neumann and Morgenstern achieve this goal by making use of extensive measurement mentioned earlier in Section 1.6. They introduced a convex combination operator $\oplus$ to combine lotteries. Take $x \in [0, 1]$. $xL_1 \oplus (1 - x)L_2 := \langle p_{3j}\rangle_{j=1}^{n}$ where $p_{3j} = xp_{1j} + (1 - x)p_{2j}$ for all j. This operator serves the same purpose as the concatenation operator in extensive measurement.

von Neumann and Morgenstern have the following three axioms:

Axiom 1 (Weak Order): $\preccurlyeq$ is a weak order.

Axiom 2 (Independence or Cancellation of Common Parts): $\oplus$ with a common gamble doesn't affect preference.

Axiom 3 (Archimedean or Continuity): if $L_1 \prec L_2 \prec L_3$, then $\exists 0 < x, y < 1$, $xL_1 \oplus (1 - x)L_3 \prec L_2 \prec yL_1 \oplus (1 - y)L_3$. Or alternatively, $\exists 0 < z < 1$, $zL_1 \oplus (1 - z)L_3 \sim L_2$.

von Neumann noted himself that the v-M result is obtained in the style of extensive measurement. One can see this clearly from the above three axioms. The purpose of these third axiom has been explained earlier in Section 1.6 for P6. Here, the first formulation, known as the Archimedean property, can be understood as that for any separation in preference, there is always a sufficiently small deviant in probability that will maintain it. It's in the same spirit as the alternative reading I give earlier for Savage's P6, that any length can be always be further simulated. A technically remark on this is that the preference over L_1, L_2, L_3 must be strict on at least one side. If one make the preference non-strict by replacing $\prec$ with $\preccurlyeq$ throughout the statement, it will not work. For we want to make sure that when preferences in the antecedent are strict, preferences in the consequence must be strict. (Check with Teddy on This.)

1.8 P7

P1-P6 gives us a theory for simple acts. Savage thinks that a theory of simple acts should be adequate for practical purposes. Non-simple acts assume infinitely many different potential outcomes and are hence non-realistic. He however suggests that it can be mathematically convenient to have a theory of all acts and proposes P7 to achieve this goal.

Formally, the utility of an act is just the integral of consequences against states. With P1-P6, we have defined utility for simple acts, that is, integral of simple functions over the state space. (Technically, Savage has only defined utility for gambles but not acts. It's permissible in Savage's theory to take the utility of the corresponding gamble of an act as the utility of that act. It won't be the case once unbounded utilities are allowed, as shown in [Seidenfeld et al., 2009]. Therefore, in the following, I will use $U(f)$ as the utility of act f without mentioning $U[f]$, which is the utility of gamble $[f]$ for Savage.) To accommodate non-simple acts, Savage has to specify a way to integrate non-simple functions.

The standard way to approximate the integration of non-simple functions by simple ones from below assumes continuity (that is, countable additivity), which is not desirable for Savage. Mathematicians have defined various different ways to do integration with respect to finitely additive measures (see [Luxemburg, 1991] for example). P7, essentially, lays down a condition for how integrals for non-simple acts should be defined, namely that the dominance principle should be respected.

P7: If $f \leq g(s)$ given B for every $s \in B$, then $f \leq g$ given B.

To illustrate P7, Savage gives an example, where P1-P6 are satisfied but P7 is not. Let the set of consequences C be the real interval $[0, 1)$. Let U be a utility function, such that for every consequence c, $U(c) = c$ and for every act f, $U(f) = E(f)$, where $E(f)$ is the expectation of f. By stipulation, U is a utility function that satisfies P1-P6. Define function

$$V(f) = \lim_{\epsilon \to 0} P\{f(s) \geq 1 - \epsilon\}.$$

Notice V is always in the integral $[0, 1]$, by definition. For every simple act f, V is always zero. Take any simple act f. $f(s)$ is bounded by a concrete real number less than 1, which is the best consequence f can bring. As $1 - \epsilon$ gets closer to 1, $1-\epsilon$ at some stages becomes larger than the upper bound of $f(s)$, which makes $P\{f(s) \geq 1-\epsilon\} = 0$ and hence $V(f) = 0$. However, V is not always 0 for non-simple acts. Therefore, intuitively, V is function that rewards non-simple acts, but not simple ones.

Define $W(f) = U(f) + V(f)$, for every act f. Define a binary relation $\preceq$ over acts by $f \preceq g$ iff $W(f) \leq W(g)$. One can check that $\preceq$ satisfies P1-P6. The reason behind is that $W(f) = U(f)$ for every simple act f. In this setting, the following act violates P7, but not P1-P6.

Take the set of all positive integers the state space S. Probability P assigns 0 to each single integer but 0.5 for the integer to be even and 0.5 for odd. Such P can be constructed explicitly. It will violate countably additivity, but not finitely additivity. Define act

$$f = \begin{cases} 0 & \text{if s is odd;} \\ 1 - 1/n & \text{if s is even.} \end{cases}$$

$V(f) = \lim_{\epsilon \to 0} P\{f(s) \geq 1-\epsilon\} = 1/2$ since for any ϵ, in all except finitely many states s that are even, $f(s) \geq 1-\epsilon$ will be satisfied. Those finitely many even states have 0 probability combined.

The first problem this example arises is that despite W is a utility function for simple acts, it fails to be a utility function for f. For W to be a utility function of f, the following should hold:

$$W(f) = (W(f)|\text{n is odd})P(\text{odd}) + (W(f)|\text{n is even})P(\text{even}).$$

However, given $P(\text{odd}) = 0$, $P(\text{even}) = 1/2$ and $(W(f)|\text{n is even})$ is bounded from above by 1, the above formula cannot hold.

Now, we show P7 is violated in this example. Take $g(s) = max\{3/4, f(s)\}$ for every s. By P2, $g \succ f$. Take any s, $W(g(s)) \leq 1 = W(f)$. By definition of $\preceq$, for every s, $g(s) \preceq f$. By P7, $g \preceq f$, which contradicts $g \succ f$.

Savage left a remark that this example roots from finite additivity. Such remark isn't quite right. The violation of P7 can be obtained in similar manner for countably additive probabilities. One simply has to use the same trick at a higher cardinal level, by defining a function that favors acts with uncountably many different outcomes.

One disadvantage of P7 pointed out by Fishburn (see Savage's footnote on p.80) is that it (together with P1-P6) implies bounded utility. (Recall that Savage's theory for simple acts (P1-P6) does precludes infinite utility by P4 the Archimedean axiom but allows unbounded utilities. The utility function can take any finite (but not infinite) value. Therefore, simple acts can have arbitrary finite utility but not infinite utility given simple acts can only have finitely many different outcomes.) This point can be shown by considering the St. Petersburg gamble as follows.

Take the set of denumerably many events $\{E_1, E_2, ...\}$ as a partition of the state space. Suppose probability $P(E_n) = 1/2^n$ and act $f(E_n) = 2^n$. We may call the act f "St. Petersburg act", since it codes the St. Petersburg paradox. It can be easily shown that $U(f) > k$ for any real number k.

Consider act $g := \frac{1}{2}0 \oplus \frac{1}{2}f$, that is, for probability 1/2, g gives you constant act 0 and for probability 1/2, g gives you act f. Similarly, act $h := \frac{1}{2}2 \oplus \frac{1}{2}f$. By P2, $h \succ g$. However, for every s, $h(s) = k_s$ for some real number k_s. Hence, $g \succeq h(s)$ for every s. By P7, $g \succeq h$.

1.8.1 Questionable Privilege of the State Space

An alternative way to read P7 is that it privileges the state space as a partition of itself with conglomerability (see Section 2.4 for definition of conglomerability). This privilege, as all other privileges in philosophy, is questionable.

On one hand, one may ask the philosophical/metaphysical question, that what makes the state space special as a partition so that it's privileged to be conglomerable while other partitions of the state space are not so privileged. On the other hand, it raises problems when one tries to extend the original state space by considering finer distinctions. Originally, the agent was deciding between pizza, burger and pasta. Later, the agent decides to take more factors into consideration and distinguish veggie burger from cheese burger, etc. This is clearly a shift on the choice of consequences, but also forces the agent to have a finer state spaces. The original state space is then a partition of this finer state space. In this finer state space, should the original state space keep its privilege of being conglomerable? Or should we start all overall again and deprive this privilege from the original state space?

Chapter 2

Some Challenges Faced by Savage's Theory

We already discussed some problems people raise against Savage's theory as we introduce Savage's postulates.

It's worthwhile to give a brief summary here. In 1.3, I discussed some motivation to deny Sen's condition α and hence Savage's P1. In 1.4, I introduced Allais paradox and Ellsberg paradox; in both paradoxes, choices found intuitive contradict P1+P2; in particular, higher-order uncertainty is introduced in Ellsberg paradox to show Savage's theory doesn't exhaust uncertainty. In 1.5, I discussed why not all deviant choices of consequence can be excluded from Savage's theory; as consequence, Savage's theory fails to generate unique qualitative or quantitative probabilities; this problem can be called the problem of state-dependent utilities, or the problem of rigid rods, or the problem of numeraire; it's discussed in further detail in 1.5.3. In 1.8, I also mentioned one reservation one may have against Savage's P7, that, together with P1-P6, it entails bounded utility.

In this section, we will discuss some further challenges Savage's theory faces.

2.1 Higher-Order Uncertainty

We already discussed Ellsberg paradox in earlier in 1.4, which is a famous early example of higher-order uncertainty. Overall, the idea of higher-order uncertainty should be intuitive.

Suppose I assign 1/2 for heads to both (fair) coin flipping and (fair) coin spinning. But I'm not as confident with my prior on coin spinning as I'm with my prior on coin flipping: I assigned 1/2 for coin flipping with certainty while

I assigned 1/2 for coin spinning only because I don't know what else should I assign. Apparently, I have a higher-order uncertainty (uncertain about my priors) regarding coin spinning.

The usual Bayesian response to higher-order uncertainty rests higher-order uncertainty on the likelihood ratio the agent has when updating on new evidence. Suppose I'm certain with my prior on coin flipping but uncertain with my prior on coin spinning. Then seeing 10 heads in a row in the case of flipping wouldn't change my priori as much it does in the case of spinning. In the case of flipping, I would regard it as a coincidence while in the case of spinning, I would perhaps regard it as some feature of coin spinning previously unknown to me.

Savage gives a different response to higher-order uncertainty. Savage argues that if we put higher order uncertainties into our theory by representing them as probabilities, the resulting theory might better represent our psychological states but doesn't answer differently to any decision problem. That is, higher probabilities do not influence our decision making at all.

Take P to be my personal probability. P(R) is my first-order personal probability on some event R, e.g. coin landing on heads. Take Q to be the random variable on P(R). P(Q), hence, is my second-order probability on my first-order probability P(R). No so surprisingly, Savage shows that $E_P(Q) = P(R)$. That is the expectation of my second-order probability over my first-order probability is just my first-order probability. This result isn't surprising. For $P(Q)$ to be my personal probability on $P(R)$, it better satisfies that $E_P(Q) = P(R)$, otherwise I'd better revise at least one of $P(Q)$ and $P(R)$, given I have introspective access and control over my personal probabilities. Savage assumes (perhaps correctly) that our higher-order probabilities cannot influence decision making directly, but can only do so via influencing first-order probabilities. First-order probabilities, however, will not get influenced by higher-order probabilities, given the result above.

Savage's argument isn't quite tight for he is making the underlying assumptions (1) that higher-order uncertainties should be treated as probabilities; and (2) that only expectations (but not say distributions) of higher-order probabilities have influence on lower ones.

I. J. Good denies Savage's assumption (1) that higher-order uncertainties should be represented as probabilities. He studied the hierarchy of higher-order uncertainties using various representations in his influential books, [Good et al., 1966] and [Good, 1983].

Ellsberg in his dissertation also surveyed several ways to solve the Ellsberg paradox, including treating higher-order uncertainties as sets of probabilities (instead of merely probabilities).

In my view, granting the assumption (1), the assumption (2) also essentially trivializes higher-order probabilities. Further investigations are needed to see if it's possible to treat higher-order uncertainty as probabilities without rendering them to triviality. (Perhaps, such higher-order probabilities, with Savage's trivialization, will simply reduce to the standard Bayesian reply introduced in the beginning of this section.)

2.2 Uncertainty in Mathematics and Logic

[Savage, 1967][1] discussed the problem of uncertainty in mathematical and logical statements. I already mentioned earlier that Savage intends his theory of exhaust all of uncertainties an agent may have. However, the term 'all' here is restricted to the domain of empirical studies that do not include mathematics and logic.

Savage's theory is built upon logic and a fair amount of mathematics (at least strong enough to prove weak/strong law of large numbers). As a consequence, a fair amount (if not all) mathematical and logical statements that follow logically from those assumed logical and mathematical facts cannot be assigned non-trivial probabilities (and hence non-trivial priories) in Savage's theory.

Without running into logical contradictions, Savage's theory has to assign all logical and mathematical statements that are entailed by itself probability 1. This makes every Savage agent logical and mathematical omniscient: for every mathematical or logical statement that is entailed by Savage's theory, every Savage agent must assign the personal probability 1 on that statement. This problem is a general problem for theories that attempt to characterize uncertainties with probabilities.

This feature of Savage's theory and Bayesian theories in general, posts troubles for Bayesian statistics/epistemology to be taken as a general scientific methodology. This point is made clear by the old evidence problem. This problem is commonly attributed to Clark Glymour but clearly known to the community for long. The following is one version of it.

Einstein in 1915 published a mathematical derivation that explains the anomalous advance of the perihelion of Mercury, which is an evidence known for long (old evidence). Einstein's theory gets much support/confirmation from this derivation and the personal probabilities people in the scientific community have over general relativity have increased significantly after seeing this mathematical derivation. To characterize this phenomenon in a Bayesian framework, we expect that the priori probability an astrophysicist has over the theory of general relativity $P(GR)$ should increase once conditional on the fact that general relativity entails the the anomalous advance of the perihelion of Mercury M. That is $P(GR) < P(GR|GR \vdash M)$. However, $GR \vdash M$ is a mathematical fact (which presumably doesn't require more resources to prove than the law of large numbers) and must have probability 1 for every personal probability

[1] The published version of this paper has an unfortunate crucial typo, which doesn't appear in his preprint.

P. However, once $P(GR \vdash M) = 1$, it's easy to see from Bayes' theorem, that it must be the case that $P(GR) = P(GR|GR \vdash M)$. Bayesian theory of confirmation, therefore, cannot explain the phenomena.

At this moment, no answer to this general problem of Bayesian has been widely endorsed. Savage originally accepts that his theory only deals with uncertainty over empirical sciences, but not mathematics. However, the old evidence example shows that it concerns physics, which is clearly an empirical science.

One apparent solution to get around this problem is to assign probability to, say, my beliefs over those mathematical statements, instead of mathematical statements directly. Savage's (or, in general, Bayesian) theory can easily accommodate that $P(B(GR)) < P(B(GR)|B(GR \vdash M)$, or better $P(GR) < P(GR|B(GR \vdash M))$, where B is the predicate for my belief, since $B(GR \vdash M)$ is not a mathematical statement.

The problem of this solution is that it disturbs the interpretation of probability. The intuitive interpretation of $P(E)$ is the agent's partial belief on the event/proposition E. $P(GR \vdash M)$ is hence the agent's partial belief on the mathematical statement $GR \vdash M$. Intuitively, agents do have partial beliefs on mathematical statements. This solution admits that such partial beliefs cannot be modeled using Bayesian methods. Rather, Bayesian methods can only model the agent's partial beliefs on his own (or others') epistemic attitudes towards mathematical/logical statements.

2.3 Conflicts between Classical and Bayesian Statistics

Savage originally intends to provide a foundation for classical statistics of his time, namely the Neyman-Pearson statistics. However, as his theory gets developed, Savage realizes that his theory conflicts with classical statistics in various ways and that he unexpectedly discovered a foundation for a different kind of statistics, now known as Bayesian statistics (though there are earlier revivals of Bayesian statistics in early 20th century, namely Harold Jeffreys' *Theory of Probability* and Wald's *Statistical Decision Functions*).

Savage's theory can be separated into two parts, namely probability and utility. The probability part alone forms a theory for inference (or confirmation), which is a version of Bayesian statistics. Probability and utility together form a theory of decision. In this section, I present several conflicts between Savage's theory of inference and classical theory of statistical inference, i.e. between Bayesian statistics and Neyman-Pearson statistics.[2]

[2] Most conflicts discussed in this section can be found in [Kadane and Seidenfeld, 1990].

2.3.1 Preliminary: Neyman-Pearson Tests

Statistical hypothesis testing is a central piece of classical statistics. In a test, we specify two hypotheses, the null hypothesis H_0, which is considered as the received view and the alternative hypothesis H_1, which will replace the null hypothesis when the null hypothesis gets rejected. In each test, we take evidence against H_0 and see if our evidence is sufficient to reject H_0. If so, we reject H_0 and replace H_0 by H_1. Otherwise, we conclude that we should not reject H_0 and continue to take H_0 as the received view. (The later case is sometimes rather misleadingly called 'to accept H_0'.)

Statistical tests are subject to two types of errors, known as Type-1 error and Type-2 error. Type-1 error, a.k.a. false positive, is the case where the test rejects H_0 when H_0 is true. Type-2 error, a.k.a. false negative, is the case where the test doesn't reject H_0 when H_0 is false.

We measure the chance of a test committing these two kinds of errors by significance and power respectively.

The significance of a test is defined as $1-\alpha$, where α is the probability of the test committing Type-1 error $P(\text{Reject } H_0|H_0 \text{ is true})$. For a test of, say, 95% significance, the probability of the test not committing false positive is 0.95, and the probability of it committing false positive is 0.05. Notice that H_1 doesn't appear in the definition of significance.

The power of a test is the probability $P(\text{Reject } H_0|H_1 \text{ is true})$. Define the probability of making a Type-2 error as $\beta = P(\text{Not reject } H_0|H_1 \text{ is true})$. The statistical power of by definition $1 - \beta$. For a test of, say, 95% power, the probability of the test not committing false negative is 0.95 and the probability of it committing false negative is 0.05. Notice that H_1 plays an essential role in the definition of power. As a consequence, the power of a test can only be defined once both H_0 and H_1 are specified. When the alternative hypothesis is not specific enough, e.g. H_1 is simply the negation of H_0, the power of a test can be hard to compute.

Neyman-Pearson hypothesis testing prioritize the avoidance of Type-1 error over the avoidance of Type-2 error. Typically, we first fix an α and hence fix the upper bound of getting a Type-1 error. Then we look more the most powerful test.

The standard procedure to run a Neyman-Pearson test is first to specify a level of significance by specifying an α, say 0.05. Then we compute the p-value $p = P(X|H_0)$, which is the probability of getting the sample data X given the null hypothesis H_0 is true. Lower p is, more likely H_0 is false. If $p < \alpha$, we have more than $1 - \alpha$ confidence to reject H_0 without committing a Type-1 error.

2.3.2 Preliminary: Sufficient, Ancillary and Suppressible Statistics

In this section, we introduce three different notions: *sufficient*, *ancillary* and *suppressible*. The first two notions are well known to modern statisticians, while the last one is proposed by [Kadane and Seidenfeld, 1990] to demarcate when some statistics is considered as relevant by Bayesian statistics.

A statistics $s = T(D)$ for data D is *sufficient* for a set of hypothesis $\mathcal{H}$, if the distribution of the data D, given the statistics s, is independent with every hypothesis $h \in \mathcal{H}$, i.e. for every $h \in \mathcal{H}$, $P(d|s, h) = P(d|s)$.

If s is sufficient, then

$$\frac{P(d|h_1)}{P(d|h_2)} = \frac{P(s|h_1)}{P(s|h_2)},$$

since

$$\frac{P(d|h_1)}{P(d|h_2)} = \frac{P(d|h_1, s) \times P(s|h_1)}{P(d|h_2, s) \times P(s|h_2)} = \frac{P(d|s) \times P(s|h_1)}{P(d|s) \times P(s|h_2)} = \frac{P(s|h_1)}{P(s|h_2)}.$$

Recall that the Bayes rule is as follows:

$$\underbrace{\frac{P(H \mid E)}{P(H' \mid E)}}_{PosteriorOdd} = \underbrace{\frac{P(E \mid H)}{P(E \mid H')}}_{LikelihoodRatio} \times \underbrace{\frac{P(H)}{P(H')}}_{PriorOdd}$$

The likelihood ratio alone is sufficient to perform a Bayesian update using the Bayes rule. A "naive" Bayesian following Savage's single agent Bayesian theory must be a so called a likelihood theorist, always taking likelihood ratios as sufficient for statistical data analysis. A statistics that is sufficient as defined above, therefore, would always be sufficient for a naive Bayesian. Classical statistics, on the other hand, sometimes takes information additional to those sufficient as defined into consideration. One such example is given later in 2.3.3.

The situation is more interesting for ancillary statistics.

A statistics t is *ancillary* if its distribution is independent of the parameter of interest θ. That is t is ancillary iff $P(t|\theta) = P(t)$.

One example of ancillary data is as follows. Suppose that we are interested in the distribution θ of flipping a particular coin. We use, say, a roulette wheel to decide the number of flips n we are going to make. We then make n flips and record the number of heads m. In this case, n is an ancillary data for θ.

It's not hard to see that an ancillary statistics by itself cannot tell us anything about θ. By the Bayes' rule,

$$P(\theta|t) = \frac{P(t|\theta)P(\theta)}{P(t)} = P(\theta).$$

But, when combined with other statistics, ancillary statistics may give information about θ. Suppose data d can be written as the pair $d = (s, t)$, with t ancillary for θ.

$$
\begin{aligned}
P(\theta|d) &= \frac{P(d|\theta)P(\theta)}{P(d)} \\
&= \frac{P(s,t|\theta)P(\theta)}{P(s,t)} \\
&= \frac{P(s|t,\theta)P(t|\theta)P(\theta)}{P(s,t)} \\
&= \frac{P(t)}{P(s,t)} \times P(s|t,\theta) \times P(\theta) \\
&= \frac{P(s|t,\theta)}{P(s|t)} \times P(\theta)
\end{aligned}
\tag{2.1}
$$

From the derivation above, we see that t provides relevant information to $P(\theta|d)$ via $P(s|t,\theta)$.

Define a statistics t to be *suppressible* for θ in the present of s iff t is ancillary for θ and $P(s|t,\theta) = P(s|\theta)$.

It can be shown that whenever t is suppressible, $P(t|s) = P(t)$, as the following:

$$
\begin{aligned}
P(s|t) &= \int_\theta P(s|t,\theta)P(\theta|t)d\theta \\
&= \int_\theta P(s|\theta)P(\theta|t)d\theta \\
&= \int_\theta P(s|\theta)P(\theta)d\theta \\
&= P(s)
\end{aligned}
\tag{2.2}
$$

Using this result, continuing the derivation above, we have

$$
\begin{aligned}
P(\theta|d) &= \frac{P(s|t,\theta)}{P(s|t)} \times P(\theta) \\
&= \frac{P(s|\theta)}{P(s)} \times P(\theta) \\
&= P(\theta|s)
\end{aligned}
\tag{2.3}
$$

This shows that s contains all the (likelihood) information about θ in d. It can be shown that s is sufficient as defined earlier:

$$\begin{aligned} P(d|s,\theta) &= P(t|s,\theta) \\ &= \frac{P(\theta|t,s)P(t|s)}{P(\theta|s)} \\ &= \frac{P(\theta|s)P(t|s)}{P(\theta|s)} \\ &= P(t|s) \\ &= P(d|s) \end{aligned} \tag{2.4}$$

Bayesian statistics and classical statistics give opposing verdicts for the relevance of certain ancillary statistics. In the cases presented in 2.3.3 and 2.3.4, Bayesian statistics judges certain ancillary statistics to be suppressible, while classical statistics judges them to be relevant. In the cases presented in 2.3.5, 2.3.6 and 2.3.7, certain ancillary statistics considered irrelevant by classical statistics are not suppressible for Bayesian.

In the following, five examples of conflicts between Bayesian and classical statistics are presented. Savage is (or would be) pretty confident with the answers his theory provides to the first three examples, but thinks the last conflict is a real problem of his theory.

2.3.3 Conflict 1: Optional Stopping

The problem of optional stopping concerns ancillary statistics.

Suppose we are interested in the distribution θ of flipping a particular coin. We have two experiments generating data of form (n, m),where n is the number of tosses and m is the number of heads, as follows:

Ex_1: Keep flipping the coin until we have 6 heads in total.

Ex_2: Flip the coin 17 times.

Suppose by coincidence, when we run the first experiment, we get the 6th heads in the 17th toss and when we run the second experiment, we get 6 heads in total. So these two experiments, by coincidence, generated the same data, namely $(17, 6)$.

For a likelihood theorist, it doesn't matter whether this data $(17, 6)$ is generated by Ex_1 or Ex_2.

However, for Neyman-Pearson tests, these two experiments give you different reject regions for the same hypothesis. For Neyman-Pearson testing, rejection regions are determined by the likelihood distribution of the entire space. Ex_1 and Ex_2 have different spaces (since for Ex_1 n can take any integer larger than or equal to 6 as value, while for Ex_2 n must be 17), and hence different likelihood distribution over data points. So for certain hypothesis, the same data from these two experiments can give you different conclusions about whether you should reject that hypothesis or not in Neyman-Pearson testing.

More generally, likelihood-theorists and Bayesian statisticians only care about what did occur, while classical statisticians care not only about what did occur but also what could have occurred. One may take this as a disadvantage of Bayesian statistics. [Schervish et al., 2004] argues that this is not. One cannot use optional stopping to “fool” a Bayesian.

In this case, Classical statistician is taking some information that is suppressible for Bayesian as relevant.

2.3.4 Conflict 2: Mixed Hypothesis Testing

As mentioned before, Neyman-Pearson testing prioritize the avoidance of Type-1 error over the avoidance of Type-2 error by design. We first fix a level of significance by picking some α and then look for the most powerful test of that significance level.

Consider the test of the simple[3] null hypothesis $H_0 : \theta = \theta_0$ against the simple alternative hypothesis $H_1 : \theta = \theta_1$. Let C be a critical regions of size α (i.e. let C be a region such that $\alpha = P(C, \theta_0) = P(D, \theta_0)$). C is **the most powerful size α test** if for every critical region D, $P(C, \theta_1) \geq P(D, \theta_1)$.

Intuitively, the most powerful size α test is the region of size α where H_1 is most likely to be true.

The Neyman-Pearson lemma shows that a likelihood ratio test is the most powerful test of a simple hypothesis against a simple hypothesis when both hypotheses have continuous support.

The Neyman-Pearson Lemma. Suppose we have a random sample $X_1, X_2, ..., X_n$ from a probability distribution with parameter θ. Then, if C is a critical region of size α and k is a constant such that:

$$\frac{L(\theta_0)}{L(\theta_1)} \leq k \text{ inside the critical region } C$$

and

$$\frac{L(\theta_0)}{L(\theta_1)} \geq k \text{ outside the critical region } C$$

then C is the most powerful critical region for testing the simple null hypothesis $H_0 : \theta = \theta_0$ and $H_1 : \theta = \theta_1$.

When we fix α for testing a discrete null-hypothesis against a discrete alternative, we can improve the power over non-mixed tests by using additional,

[3]When a random sample is taken from a distribution with parameter θ, a hypothesis is **simple** if the hypothesis uniquely specifies the distribution of the population from which the sample is taken.

randomized data to simulate continuity in a mixed-test and achieve the most powerful test.

The following is a rather dramatical example. Suppose we have some experiment generating data s in the following way.

	$s=0$	$s=1$	$s=2$	...	$s=10$
$P(s\|h_0)$	0.05	$\frac{0.95}{55}*1 \approx 0.01727$	$\frac{0.95}{55}*2 \approx 0.03454$	...	$\frac{0.95}{55}*10 \approx 0.1727$
$P(s\|h_1)$	0.5	$\frac{0.5}{55}*1 \approx 0.00909$	$\frac{0.5}{55}*2 \approx 0.01818$	...	$\frac{0.5}{55}*10 \approx 0.0909$

Table 2.1: An experiment for discrete parameter s

If we fix $\alpha \leq 0.02$, the only non-mixed test of h_0 based on datum s is to reject H_0 when $s = 1$. When $s \neq 1$, the probability of making Type-1 error $P(\text{Reject } H_0|H_0 \text{ is false})$ is more than 0.02. This non-mixed test in this case is a biased test. When $s = 1$, it's more likely for H_0 to be true (with probability $\frac{0.017}{0.017+0.009}$) than for H_1 to be true (with probability $\frac{0.009}{0.017+0.009}$). That is, the power of this test is less than its significance.

However, with the addition of a uniformly distributed random number $t \in \{0, 1, ..., 9\}$. We can chop up the case of $s = 0$ into 5 pieces, and run the most powerful mixed-test for $\alpha = 0.02$ that rejects h_0 when $s = 0$ and $t \in \{0, 1, 2, 3\}$.

By the Neyman-Pearson lemma, this mixed test is the most powerful test for $\{H_0, H_1\}$. However, it violates the Bayesian theory. For t is clearly ancillary for $\{h_0, h_1\}$ and suppressible in the presence of s, given $P(s|t, h_0) = P(s|h_0)$ and $P(s|t, h_1) = P(s|h_1)$.

This reasoning for Bayesian works in general. There is no Bayesian justification for using mixed-tests, which is sometimes recommended by Neyman-Pearson lemma. However, this point is mainly a scholastic point. Classical statisticians typically do not want to run mixed tests, for it makes the testing result relying on some random events. In presence of examples like the one above, they typically pursue other experiments.

In this conflict, against certain information/statistics that is suppressible for Bayesian is considered relevant by classical statistics.

2.3.5 Conflict 3: Attitudes towards Ancillary Statistics from Random Sampling

This section gives some motivations for taking some ancillary statistics into consideration, by providing a non-realistic toy example. A more realistic one will be given in next section.

Suppose we are interested in the political opinions θ (say, the proportion of Democrats supporters) in the population. We conduct a poll by a random sampling and gather data (s, g), where s is the proportion of Democrats supporters in the sample and g is the gender distribution of the sample.

Suppose our sampling procedure is independent with the interested parameter θ. Then $P(g) = P(g|\theta)$ and g is therefore ancillary.

For classical statistics, given g is randomly generated, when we try to draw conclusions about the parameter θ in the population (including both male and female), the statistics g is irrelevant and shouldn't be taken into consideration.

For Savage or Bayesian in general, once we have the distribution of g, we would condition on g regardless and given g correlates with θ, it will make a difference to our posterior. Intuitively, $P(\theta|s)$ shouldn't be equal to $P(\theta|s, g)$. As the gender distribution of my sample is skewed towards, say, male, I should take such skewness into consideration when updating. That is, g is ancillary but not suppressible.

The Bayesian treatment is especially compelling when the sample gender distribution is radically different from the population gender distribution. The overall idea is that once the randomization is done and some information has been generated from the random process, we should take all information, including those resulting from the randomization, into consideration. It's expressed by Savage in [Savage, 1962], pp. 87 - 91,[4] using examples from Fisher about Latin squares and Knut Vik squares.

Latin Square is a certain kind of randomization used to allocate crops with different treatments over non-uniform soils. It's studied extensively in Fisher's seminal book *The Design of Experiments* among other places. Knut Vik square is a particular kind of Latin square that makes the allocation of crops with different treatments particularly "diverse". Fisher argues against Knut Vik (designed diversity) in favor of Latin Squares (randomness). Suppose we have five treatments on crops and we are interested in the difference among the effects of them. Suppose that the soil is not uniform and is divided into 5 by 5 squares. Latin Square requires that each treatment only appears once in every row and every column. Knut Vik square further requires that each two block of the same treatment should be separated by a Knight move in chess. Fisher argues that if we randomly choose a Latin Square for our experiment, we can learn the difference of soil among each row and among each column by comparing the row average of each row and the column average of each column. We can then remove the difference among rows and difference among columns and get more accurate variance for the treatments. Knut Vik square, on the other hand, maximizes the variance within each treatment. Given we are interested in each treatment, we are interested in the variance of each treatment from the total average. The variance of each treatment from the total average has two parts: one part from the variance within the treatment, another part from the variance between the treatment and other treatments. We are interested in the second one, but not the first one. However, it's the first one that is maximized by Knut Vik square. As a result, Knut Vik square makes the comparison among

[4] It starts from the bottom of p. 87 with Cox's question

treatments hardier, compared to Latin Square.

In [Savage, 1962], p. 88, Savage mentioned his earlier conversation with Fisher, where he asked Fisher when he randomly chooses a Latin square but ends up with a Knut Vik square, what should he do. Fisher answers that he should do the random selection again. This example of Latin square and Knut Vik square is not different from the toy example I give earlier in this section. Savage says that this suggests that such criticism that the result of the randomization should be taken into consideration has been around even before Bayesian statistics.

2.3.6 Conflict 4: Randomization in Data Analysis

This section gives a more realistic example of non-suppressible ancillary statistics and in particular, the usage of randomization in data analysis (i.e. randomize after we get the data).

The Behrens-Fisher problem is a famous problem that separates Bayesian and Neyman-Pearson statistics. It was first used by Fisher as an example to distinguish his own statistical methods from Neyman-Pearson (although Neyman-Pearson statistics was developed from Fisher's work).

Suppose that $x_1, ...x_n$ are i.i.d. draws from normal distribution $N(\mu_x, \sigma_x^2)$ and $y_1, ...y_n$ are i.i.d. draws from normal distribution $N(\mu_y, \sigma_y^2)$. The parameter of interest is $\delta = \mu_x - \mu_y$.

The rejection region for testing δ cannot be defined without knowing the nuisance parameter $v := \frac{\sigma_x^2}{\sigma_y^2}$[5]. However, v is not specified in the problem and hence, no Neyman-Pearson testing can be done directly.

A Bayesian analysis can be easily done for this problem, since both $P(\delta|\bar{x}, \bar{y}, v, s_x^2, s_y^2)$ and $P(v|s_x^2, s_y^2)$ are well-defined, where $\bar{x}$ and $\bar{y}$ are sample means, s_x^2 and s_y^2 are sample variance respectively. Then $P(\delta|\bar{x}, \bar{y}, s_x^2, s_y^2) = \int_v P(\delta|\bar{x}, \bar{y}, s_x^2, s_y^2, v)P(v|s_x^2, s_y^2)dv$.

From a Bayesian perspective, Neyman-Pearson statistics can be partially recovered in Bayesian statistics by using certain bridging priors. This is not the case for the Behrens-Fisher problem. Although both $P(\delta|\bar{x}, \bar{y}, s_x^2, s_y^2, v)$ and $P(v|s_x^2, s_y^2)$ have confidence interval interpretation, their product $P(\delta|\bar{x}, \bar{y}, s_x^2, s_y^2)$ doesn't have one.

Fisher provided his own solution to this problem and criticize Neyman-Pearson methods for not being able to solve this problem. Harold Jeffreys, later, in his *Theory of Probability* argues that what Fisher did to the Behrens-Fisher problem is essentially moving towards Bayesian. For Fisher did calculate $P(\delta|\bar{x}, \bar{y}, s_x^2, s_y^2, v)$ and $P(v|s_x^2, s_y^2)$ and calculate their product as a Bayesian would, except that Fisher interprets the result different from Bayesian or Neyman-

[5] A parameter is nuisance if it's not of immediate interest but must be accounted for in the analysis of those parameters of interest

Pearson.

By matching $X's$ and $Y's$ randomly to form n pairs (x_i, y_i), define $z_i = x_i - y_i$. z_i's are then i.i.d. draws from $N(\delta, \sigma_x^2 + sigma_y^2)$. An exact answer to the Behrens-Fisher problem can be given by applying t-test (which doesn't require to known variance), independent of the nuisance factor $\frac{\sigma_x}{\sigma_y}$, on z_i.

This treatment explicitly uses randomization in data analysis. Savage and Bayesian in general do not like it. One generic objection to this treatment is that it throws out certain data. Only the difference between x_i and y_i are used while, for example, their sum, which is known to us, isn't used. A more Bayesian objection is that certain non-suppressible ancillary statistics is ignored, i.e. treated as suppressible, in this practice.

Let $s = \{(x_i, y_i)\}$ and let $t =$ the rank order (out of n!) of the sample variance s_z^2 created by the permutation of subscripts used to form z_i. Then, t is ancillary for δ under the random pairings. However, t is not suppressible, as $P(s|\delta, t) \neq P(s|\delta)$. Specifically, $P(s|\delta, t)$ but not $P(s|\delta)$ involves the nuisance factor $\frac{\delta_x}{\delta_y}$.

2.3.7 Conflict 5: Randomization in Experimental Design

Different from his attitude towards randomization in data analysis, Savage thinks that randomization does play an important role in experimental design. However, as we will show later, Savage theory implies that randomization is useless in experimental design. This is taken by Savage as a genuine defect of his theory and moves him together the theory of multi-agent decision making.

In [Savage, 1961], page 585, Savage says: "Applying the theory (of personal probability) naively one quickly comes to the conclusion that randomization is without value for statistics. This conclusion does not sound right; and it is not right. Closer examination of the road to this untenable conclusion does lead to new insights into the role and limitations of randomization but does by no means deprive randomization of its important function in statistics."

Ronald Fisher proposed and defended the usage of randomization in experimental design in his seminal book *The Design of Experiments*. His book starts with the "Lady Tasting Tea" example to illustrate the importance of randomization in experimental design.

A lady claims that she can tell when a cup to tea with milk is made by putting milk in first or by putting tea in first. To test if the lady has such skill, we take eight cups of mixtures of tea and milk, including 4 tea-first (i.e. first put in tea then milk) and 4 milk-first. After having controls over irrelevant variables, e.g. temperature, volume of tea and milk, size of cups etc., Fisher argues that there are still reasons to assign tea-first's and milk-first's randomly to cups.

(1) By assigning randomly, we are (supposed) to be assured that her success

is due to her skill of tasting the drink and not due to, e.g. her ability for guessing a non-randomly chosen order fixed by the experimenter.

(2) By assigning randomly, the study is (supposed) to be assured against confounding of uncontrolled factors. There could be some unnoticed common to the experimenter putting tea-first in some particular cups and to the lady guessing those cups are tea-first, e.g. some feature of the cups that are not explicitly recognized by the experimenter but nevertheless affects the choice of the experimenter. In this case, randomization works like an intervention.

(3) By assigning randomly, weare (supposed to be) warranted in adopting a particular statistical model governing experimental outcomes under the null hypothesis that the Lady has no skill. In this case, the null hypothesis is just the equi-probable (1/70) distribution for possible guesses of 4 out of 8.

These three reasons are commonly found compelling (even by Savage). However, none of these three reasons can be given according to Savage's theory. Once we assign the drinks randomly, we come to know the specific order of the drinks. That order can be shown to be ancillary but not suppressible. Once we condition on the result we get from the experiment together with the information about the specific order of the drinks, all three reasons above cannot be given anymore.

Savage is aware of this problem and tentatively suggests that the need of randomization comes from the theory of group decision.

"Though we all feel sure that randomization is an important invention, the theory of subjective probability reminds us that we have not fully understood randomization... The need for randomization presumably lies in the imperfection of actual people and, perhaps, in the fact that more than one person is ordinarily concerned with an investigation." [Savage, 1962]

[Kadane and Seidenfeld, 1990] pursues some development of this idea (that the need of randomization lies in the fact that more than one person is ordinarily concerned with an investigation) and proposes the distinction between experiments to learn (single agent) and experiments to prove (multi-agents).

2.4 Non-Conglomerability of Finite Additive Probability

Non-conglomerability is a problem faced by finite additive probability in general. Savage's theory, proposed as a general theory of probability, only assumes finite additivity, which makes non-conglomerability a problem for Savage as well. This section explains what non-conglomerability is.

A probability P is **conglomerable** iff for any partition $\{X_i\}$ of the space, for any event A, if for all X_i, $c \leq P(A|X_i) \leq d$, then $c \leq P(A) \leq d$. Finite additive probabilities, in general, are not conglomerable. Indeed, it's shown in

[Schervish et al., 1984] that for every strictly finite-additive probability, there is an event and denumerable partition, for which conglomerability fails.

Non-conglomerability is one of the most concerning facts about finitely additive probability. There is no easy solution to this problem. It's a mathematical fact that finite additive probability admits the feature of non-conglomerability. Countable additive probability has its own problems. [To be added to the notes.] It's up to us to choose which to use. For example, Seidenfeld supports finite additive probability in general; Easwaran, on the other hand, thinks conglomerability is a desirable feature of our probability/credence/partial belief.

2.4.1 Dubins' example

We have two methods to generate a random natural number, A and B.

Method A generates a random natural number N by keeping tossing a fair coin until it lands heads. If the fair coin lands heads on the nth toss, it returns the natural number n. So $P(N = n|A) = 1/2^n$.

Method B generates a random positive natural number by, so to speak, rolling an infinite-face die. [6] Consequently, $P(N = n|B) = 0$ for any n. It may be worth noting that this setting is compatible with $\dfrac{P(N = n + 1|B)}{P(N = n|B)} = 1/2$, which isn't unspecified in the example.

We can assign arbitrary value to $P(A)$ and $P(B)$. Say, let $P(A) = \epsilon$ and $P(B) = 1 - \epsilon$ for some fixed small ϵ.

	$N = 1$	$N = 2$	$N = 3$	...	N=n	...
A (with $P(A) = \epsilon$)	1/2	1/4	1/8	...	$1/2^n$	...
B (with $P(B) = 1 - \epsilon$)	0	0	0	...	0	...

Table 2.2: Dubins' example

For any n, $P(A|N = n) = 1$ and $P(B|N = n) = 0$. So $1 \leq P(A|N = n) \leq 1$ for any n despite $P(A) = \epsilon$, which violates Conglomerability.

Non-conglomerability is a major issue faced by finite-additive probability. Suppose facing a random number generator, generating random positive integers in one of the two methods A and B, one wants to obtain more information about the random generator. Then actually running the random generator will give one ultimate misleading information. Once one runs the generator and observe what number is generated, by the Bayes rule, the posterior probability P' would be that $P'(A) = 1$ and $P'(B) = 0$, no matter what number is

[6] By the way, I would deny B as a legitimate mechanism. But Seidenfeld would ask questions I cannot answer, like: What do you mean by mechanism? How do you demarcate legitimate ones from illegitimate ones? Isn't method A already unrealistic enough, since you may die at some point?

actually generated. Once your probability reaches 1 or 0, Bayesian update will no longer change it anymore. So actually running the random generator would be a really bad experiment to run for learning which method is being used. [Schervish et al., 1984] shows that such bad experiments exist for every finitely additive probability. It's known that every finitely additive probability can be decomposed into a convex combination of a countably additive probability and a strictly finitely additive probability, i.e. $P = \alpha P_\sigma + \beta P_D$, where $\alpha + \beta = 1$ and P, P_σ, P_D are finitely, countably, strictly finitely additive respectively. The maximal difference between $P(E)$ and $P(E|h_i)$ is determined by β that $sup|P(E) - P(E|h_i)| = \beta$.

Non-conglomerability isn't a problem for σ-additive probability. In this example, the probability P violates the countable additivity. Once we have σ-additivity, we can sum $P(N = n|B)$ for all n and get a contradiction that $\sum_n P(N = n|B) = 0 \neq 1 = P(B)$ which contradicts the law of total probability that tells us $\sum_n P(N = n|B) = P(B)$. But P nonetheless satisfies all axioms to be a finitely additive probability.

Non-conglomerability doesn't appear if Savage's theory is limited to simple acts (P1-P6), where only finite partitions of the state space are allowed. However, once Savage attempts to accommodate non-simple acts using P7 in his theory, both the problem of non-conglomerability and the problem of conditional probability (see Section 1.4.3). Savage himself possibly doesn't take non-conglomerability as a problem for his theory, but rather a mere mathematical fact.

2.5 Sequential Decisions and Experiment Selection

Savage's theory is a static theory. It doesn't have an update operation explicitly defined and cannot handle sequential decisions. This section illustrates one aspect of this shortcoming using Savage's treatment of experiment selection as a decision problem. Dubins' example of non-conglomerability is used to illustrate the main ideas.

Recall that there are two ways of generating random natural numbers:

Method A generates a random natural number N by keeping tossing a fair coin until it lands heads. If the fair coin lands heads on the nth toss, it returns the natural number n. $P(N = n|A) = 1/2^n$. Method B generates a random positive natural number by rolling an infinite-face die. $P(N = n|B) = 0$ for any n.

	$N = 1$	$N = 2$	$N = 3$	...	N=n	...
A	1/2	1/4	1/8	...	$1/2^n$	...
B	0	0	0	...	0	...

Construct a gamble as follow: Suppose we have a random natural number generator, generating random numbers using one of the two methods A and B. One can bet on which method is used, as indicated in the following chart. If one chooses act f, one will get 0.5 if A is the case, and 0 if B is the case. Similarly, if one chooses act g, one will get 1 if B is the case, and 0 if A is the case.

	A	B
f	0.5	0
g	0	1

Suppose I have prior probability P such that $P(A) = P(B) = 1/2$. I'm allowed to run some experiments before I make my decision on the gamble. One potential experiment is to run the random number generator and see what number is generated by the random number generator. This experiment is a very special experiment. As shown in Section 2.4.1, for any n, $P(A|N = n) = 1$ and $P(B|N = n) = 0$. No matter what the result of the experiment is, my posterior probability P' will be such that $P'(A) = 1$ and $P'(B) = 0$.

Now whether I should run this experiment before I decide can be characterized as a decision problem. It's a decision problem with two acts (choices), namely Decide Now and Run Experiment. Based on my current probability P, if I decide now, I will choose act g and with probability 1/2, I will get 1 unit of utility as reward. So the utility of Decide Now is 1/2. If I choose Run Experiment, after running the experiment, I will decide to take act f, which according to my current probability P, has utility 1/4. According to my current probability, Decide Now has a higher utility than Run Experiment and therefore, I should decide now.

This example, of course, is peculiar in nature. Savage shows that if we only allow finite partitions (and hence preclude all problems raised by non-conglomerability), running experiment always has non-negative outcome. That is it's more or equally preferable to run experiments than to decide, if it takes no cost to run experiments. One should make decision when to run experiment is equally preferred as to decide now. Formally, it is when $E(Y) = E(E(Y|X))$, where Y is the gamble you are trying to decide and X is the experiment you may choose to run.

One shortcoming of Savage's theory, that can be seen in light of this section, is that it doesn't apply to sequential decisions. Suppose you are playing a sequential game. At each stage of the game, you can choose if you want to keep the money and leave or to enter the next stage where with probability 1/2, you will end up with nothing and with probability 1/2 you will get more than 2 times of your current gain. This sequential game is analogue to the decision problem whether you should decide now or run one more experiment. Per Savage, you should always choose to play one more run, for it always has more utility. However, we know that if you always choose to play one more round, you will end up with nothing for sure.

Chapter 3

Convergence and Consensus

3.1 Savage

Savage suggests that we can achieve consensus and certainty using only personal probabilities in light of shared evidence. No matter how different prior probabilities among agents are on a given issue, accumulations in shared evidence will eventually and always "wash out" the prior. It's worth mentioning that Savage's only discussing consensus and certainty in the domain of empirical inquiry, but not in the domain of mathematics or logic, since probability theory presumes mathematics and logic.

Savage intends to use the Bayes rule of conditional probability as a rule of learning. However, he fails to make this explicit in his theory. His theory, apparently, is a static theory, that generates an agent's personal probabilities by taking the agent's static unconditional preferences as input.

The Bayes rule of conditional probability says that the posterior odd between two events H and H' equals the likelihood ratio in light of evidence E times the prior odd of these two events.

$$\underbrace{\frac{P(H \mid E)}{P(H' \mid E)}}_{PosteriorOdd} = \underbrace{\frac{P(E \mid H)}{P(E \mid H')}}_{LikelihoodRatio} \times \underbrace{\frac{P(H)}{P(H')}}_{PriorOdd}$$

Using this rule of learning, Savage obtains a weak law result from some fairly strong assumptions. Savage assumes that given an empirical inquiry, all agents share the same set of hypothesis, but have different non-extreme (neither 0 nor 1) personal probabilities on each hypothesis in the set. Empirical evidences comes as independent identical distributed random variables. And for any two hypotheses H and H' and any evidence E, all agents agree on the likelihood ratio $\frac{P(E|H)}{P(E|H')}$.

By updating on an unbounded finite sequence of i.i.d. evidences, all agents' personal probabilities over this matter converge to the same distribution in probability (i.e. consensus), and suppose the inquiry is to understand some random variable, all agents' personal probabilities will converge to the characteristic function of that random variable (i.e. converge to truth). The reason behind, roughly, is that the likelihood of the true hypothesis over any other hypothesis can get arbitrarily large and "wash out" any prior.

Savage's result is not satisfying for various reasons but it is (pretty much) as strong as one can get without making further assumptions. One advantage of Savage's result is that Savage can show that the posterior probabilities converge in a reasonable amount of time, since evidences are assumed to be i.i.d. (In contrast, in Blackwell& Dubins' more general strong law result, we have no control whatsoever over the convergence rate.) But such assumption means that we are running the same experiment repeatedly, which is far from what is done in real scientific practices.

3.2 Blackwell & Dubins

[Blackwell and Dubins, 1962] provides a strong law result of convergence and consensus assuming countable additive probability. In contrast to Savage, Blackwell and Dubins do not require evidences to be independent or identical distributed. For this reason, Blackwell and Dubins cannot guarantee that convergence and consensus can be reached in a timely manner.

Blackwell and Dubins treat experiments/evidences as a sequence of random variables $\{X_1, X_2, X_3, ...\}$. Agents start with a prior probability over all these experiments. The only assumption Blackwell and Dubins impose on such probabilities is that every agent must give extreme priors (0 or 1) to the same events. That is for any two agents with prior probabilities P and Q, $P(E) = 0$ iff $Q(E) = 0$ for every event E. Agents learn from earlier experiments $X_1, X_2, ...X_m$ and make predications on later experiments $X_{m+1}, X_{m+2}, ...$

Let F^n be the future at stage n. $F^n = \{X_{n+1}, X_{n+2}, ...\}$. Let $P^n(\cdot) := P(\cdot|X_1, X_2, ..., X_n)$ and $Q^n(\cdot) := Q(\cdot|X_1, X_2, ..., X_n)$. Blackwell and Dubins use $\sup |P^n(F) - Q^n(F)|$ for $F \in F^n$ to measure the difference between posterior probabilities of P and Q. They show that with probability (either P or Q) 1, $\lim_{n\to\infty}(P^n(F) - Q^n(F)$ for every $F \in F^n$. This result gives Blackwell and Dubins consensus among agents.

The convergence result, assuming countable additivity, was achieved earlier than Blackwell and Dubins' consensus result. Take any sequence of events ω. For any event E, either E is in the sequence or E is not. After learning, the posterior probability converges to 1 if $E \in \omega$ and to 0 if $E \notin \omega$. That is

$$P(E|x_1, ..., x_n, ...) = \begin{cases} 1 & \text{if} E \in \omega; \\ 0 & \text{if} E \notin \omega. \end{cases}$$

It's worth noting that the convergence result fails for finitely additive probabilities, which, again, is shown by Dubins' example of non-conglomerability.

[I'm not sure if I described the convergence result correctly. Ask Teddy.]

3.3 de Finetti on exchangeability

De Finetti's result (1931) on convergence precedes Savage (1954) in time. At that time, de Finetti was concerned that the idea of i.i.d. takes objective chances as an underlying assumption. He wants to recover results using i.i.d. from a subjectivist's perspective, using exchangability, which is a symmetry condition on infinite sequences of random variables. The main idea is that if your prior on an infinite sequence of random variables is exchangeable, you will behave as if you are updating on i.i.d. random variables.

Consider a binary sequence $\mathcal{X} = \{X_i\}_{i=1}^{\infty}$ where $X_i \in \{0,1\}$. P is a probability on the σ-algebra of this sequence of random variables.

Probability P is 1-exchangeable if $P(X_i = 1) = p$ for every i. Intuitively, 1-exchangability requires your prior on any one element of $\mathcal{X}$ should be exchangeable with any other one element of $\mathcal{X}$.

Probability P is 2-exchangeable if your joint distribution on any two random variables X_i and X_j, where $i \neq j$ is invariant to your choice of i and j. In the case of a binary sequence, it means that $P(X_i = x_i, X_j = x_j)$ for $x_i, x_j \in \{0,1\}$ is invariant to your choice of i and j so long as $i \neq j$.

In general, probability P is k-exchangeable if your joint distribution on any k random variables $X_{i_1}, X_{i_2}, ...X_{i_k}$ (for distinct $i_1, i_2, ..., i_k$) is invariant to the choice of $i_1, i_2, ..., i_k$. That is $P(X_{i_1} = x_{i_1}, X_{i_2} = x_{i_2}, ...X_{i_k} = x_{i_k})$ for $x_{i_1}, x_{i_2}, ..., x_{i_k} \in \{0,1\}$ is invariant to your choice of $i_1, i_2, ..., i_k$ so long as they are pairwise distinct.

Given $n \leq m$, m-exchangeability implies n-exchangeability, but not vice versa. The following is an (extreme) example of 1-exchangeable but not 2-exchangeable probability. Suppose I toss a fair coin and then keep flipping it (without tossing). The sequence generated in this way is either $(0,1,0,1,...)$ or $(1,0,1,0,...)$. Based on this mechanism, I reasonably form a probability P. For any i, $P(X_i) = 1/2$, which makes P 1-exchangeable. But the joint distribution over X_i and X_j will depend on whether $i \cong j(mod2)$ or not. For $i \cong j(mod2)$, $P(X_i = X_j) = 1$ and $P(X_i \neq X_j) = 0$, while for $i \ncong j(mod2)$, $P(X_i = X_j) = 0$ and $P(X_i \neq X_j) = 1$.

We say P is exchangeable if it's k-exchangeable for every positive integer k. Overall, exchangeability implies independence or positive correlations among random variables, where positive correlation means $P(X_j = 1|X_k = 1) > P(X_j = 1)$ and negative correlation means $P(X_j = 1|X_k = 1) < P(X_j = 1)$ (as a consequence of Cauchy–Schwarz inequality: to be further detailed). Hence, exchangeability is weaker than independence. By definition, every 1-exchangeable sequence is identically distributed. But being exchangeable doesn't guarantee a distribution P to be an i.i.d., while any i.i.d. is exchangeable.

De Finetti's shows that all exchangeable binary sequences are mixtures of Bernoulli sequences: An infinite binary sequence $X_1, ..., X_n, ...$ is exchangeable if and only if there exists a unique probability measure μ on $[0, 1]$ such that for each fixed sequence of zeros and ones $(x_i)_{i=1}^n$, we have

$$P(X_1 = x_1, ..., X_n = x_n) = \int_0^1 \theta^k (1-\theta)^k d\mu(\theta),$$

where $k = \sum_{i=1}^n x_i$ is the number of 1's in $(x_i)_{i=1}^n$.

It further holds that μ is the probability measure of the limiting frequency, that is, $Y = \bar{X}^\infty = \lim_{n\to\infty} \sum_i X_i/n$. We have $P(Y \leq y) = \mu(y)$.

The Bernoulli distribution is obtained by conditioning with $Y = \theta$:

$$P(X_1 = x_1, ..., X_n = x_n | Y = \theta) = \theta^k (1-\theta)^{n-k}.$$

(More to be added about Hewitt-Savage's generalization.)

Exchangeable sequences of binary random variables can be generated by drawing marbles from an urn. Draw with replacement will result in i.i.d. sequence, while draw without replacement will generate (finite) exchangeable sequences (of hypergeometric distribution). To generate k-exchangeable but not k+1 exchangeable sequences, we need an urn with at least k marbles.

By mixing several urns, we can mimic draw with replacement, i.e. exchangeable sequences can mimic i.i.d. (More details are needed.)

It's worth noting that Savage's theory doesn't imply personal probabilities are exchangeable.

Partial exchangeability is defined by restricting the permutations to be within certain partition. For example, suppose our observation comes with two traits: trait X and gender. Suppose there is some correlation between X and gender that makes our sequence not exchangeable. We can condition on gender by restricting the permutations to happen only within each gender group. After conditioning, the sequence is exchangeable relative to X.

Chapter 4

De Finetti's on Probability

The kind of foundations Savage gives to Statistics is now known as behavioral foundations, as it tries to reduce probabilities (and utility in Savage's case) to behavioral dispositions, namely preferences.

Much earlier than Savage, [De Finetti, 1937] provides a behavior foundation for probability (while taking utilities as granted), where probability is defined using the 0-sum *Prevision game*. Later, [De Finetti, 1974] uses the one-player *Forecasting game* to overcome some problem the Prevision game may have. In this section, I present these two different though similar approaches to define probability given by de Finetti.

4.1 The Prevision Game

Let $\Omega = \{\omega_1, ..., \omega_n, ...\}$ be a countable partition of the sure event, that is, a finite or countably infinite set of states.

Let $\mathcal{X} = \{X_i : \Omega \to \mathbb{R} | i = 1, 2, ...\}$ be a countable set of (bounded) real-valued random variables defined on Ω. In another word, $\sqcup]\S\sqcup$ is a countable set of gambles. Each X_i in $\mathcal{X}$ is a gamble that gives $X_i(\omega_j) = r_{ij}$ utility as outcome in state ω_j. The value of every gamble $X_i \in \mathcal{X}$ is bounded in the sense that $-\infty < \inf_{\omega \in \Omega} X(\omega) \leq \sup_{\omega \in \Omega} X(\omega) < \infty$.

There are two players in the Prevision game: the Bookie and the Gambler.

For each random variable $X \in \mathcal{X}$, the Bookie announces a prevision (i.e. a fair price), $P(X)$, for which he's willing to buy/sell any units of X. We denote the set of previsions the Bookie gives for $\mathcal{X}$ by $P(\mathcal{X})$.

The Gambler, on the other hand, is allowed to trade (buy or sell) arbitrary amount of finitely many variables with the Bookie at the Bookie's announced prices.

For a random variable X, the Gambler can make a contract with the Bookie by specifying a real number α_X, so that the Bookie must buy[1] α_X amount of X from the Gambler at the Bookie's prevision $P(X)$. The Gambler can choose finitely many random variables and make finitely many contracts with the Bookie. Formally, we consider the Gambler as making a contract α_X for every $X \in \mathcal{X}$, while only finitely many contracts are non-zero. I.e. the Gambler specifies a list $\alpha_{\mathcal{X}} = (\alpha_X)_{X\in\mathcal{X}}$ and all except finitely many members of this list are zero.

In each state ω, each contract α_X would have an outcome for the Bookie, whose value depends on the state ω, the random variable X, Bookie's prevision $P(X)$ and the contract α_X. Note this outcome by $O_\omega(X, P(X), \alpha_X)$. It's easy to see that $O_\omega(X, P(X), \alpha_X) = \alpha_X[X(\omega) - P(X)]$.

Then the Bookie's net outcome in state ω given $\alpha_{\mathcal{X}}$ and $P(\mathcal{X})$, noted by $O(\omega, P(\mathcal{X}), \alpha_{\mathcal{X}})$ is as follows:

$$O(\omega, P(\mathcal{X}), \alpha_{\mathcal{X}}) = \sum_{X\in\mathcal{X}} O_\omega(X, P(X), \alpha_X) = \sum_{X\in\mathcal{X}} \alpha_X[X(\omega) - P(X)]$$

The Bookie's prevision $P(\mathcal{X}) = \{P(X)|X \in \mathcal{X}\}$ is *coherent*$_1$ iff there is no strategy for the Gambler that results in a sure net loss for the Bookie, i.e. there is no $\alpha_{\mathcal{X}}$ such that $\forall\omega \in \Omega, O(\omega, P(\mathcal{X}), \alpha_{\mathcal{X}}) < 0$.

Otherwise, the Bookie's prevision is *incoherent*$_1$. If the Bookie's prevision is *incoherent*$_1$, the Bookie's outcome $O(\omega, P(\mathcal{X}), \alpha_{\mathcal{X}})$ is strictly dominated by *Abstain* (i.e. making no contract at all) which has value 0 for every $\omega \in \Omega$. In another word, being *incoherent*$_1$ is by definition, subject to Dutch-booking.

De Finetti shows that a set of previsions $P(\mathcal{X})$ is *coherent*$_1$ if and only if there exists a (finitely additive) probability $\mathbf{P}$ such that each prevision is the expected values of the corresponding variables with respect to $\mathbf{P}$, i.e. $\forall X \in \mathcal{X}, E_{\mathbf{P}}[X] = P(X)$.

Applying this result of de Finetti gives us a behavioral definition of probability. Notice that $\mathcal{X}$ can be any set of random variables. Given a set of events $\mathbf{E} = \{E_1, ..., E_n, ...\}$, let $\mathcal{X} = \{X_1, ..., X_n, ...\}$ such that each X_i is the 0-1 indicator function of E_i, i.e. $\forall i, X_i(\omega) = 1$ iff $\omega \in E_i$ and $X_i(\omega) = 0$ iff $\omega \notin E_i$. It follows from de Finetti's result that there is a unique finite additive probability distribution $\mathbf{P}$ over $\mathcal{X}$, such that $P(X) = E_{\mathbf{P}}(X) = 1{*}\mathbf{P}(E_i) + 0{*}\mathbf{P}(E_i) = \mathbf{P}(E_i)$. Therefore, there is a probability $\mathbf{P}$ that agrees with the prevision$P$ over every event E, i.e. $\mathbf{P}(E) = P(E)$ for all E. De Finetti, thereby, reduces one's probability over any set of events $\mathbf{E}$ to one's previsions over corresponding 0-1 indicator functions $\mathcal{X}$ as a Bookie in the Prevision game.

This definition is subject to two challenges. First, de Finetti takes utilities as granted. If we were to operationalize this definition of probability, we have

[1]When α_X is negative, the Bookies must sell $|\alpha_X|$ amount of X to the Gambler at price $P(X)$.

to choose a unit of consequence, which leads us to the problem of choosing numeraire discussed in 1.5.3 for Savage's theory. Choosing different units of consequence will give us different probability.

Besides, it faces the problem of betting against experts. When you play the Prevision game with an expert, you may assign your previsions strategically to maximize expected utility instead of assigning according to your belief.

Suppose the Bookie has to play the Prevision game with the Gambler over the question whether the Gambler's mother's birth is on an odd day or an even day. Suppose the Gambler knowns the answer of the question, i.e. the Gambler is an expert on this question. The Bookie believe that 52% days of a year are odd and 48% are even. Suppose the Gambler is as ignorant about the answer as the Bookie is, the Bookies would price 0.52 for odd and 0.48 for even. Such previsions are $coherent_1$ as defined by de Finetti. However, when the Gambler is an expert on this question, such previsions will make the Bookie a sure loser in a different sense: the Gambler can sell the Bookie arbitrary amount of the wrong answer. Suppose Gambler's mother's birthday is actually on an odd day. Given Gambler knows that 'even' in fact has zero utility, he can sell the Bookie any amount α_{even} of 'even' and in fact make the Bookie suffer $0.48\alpha_{even}$ loss. Facing this situation, the Bookie, who knows that the Gambler knows the answer, knows that if he plays the game sincerely (according to his own belief), he would lose all of his money. He would then strategically place all bets on one side he takes to be promising. In this case, the Bookie would strategically price 'odd' for 1 and 'even' for 0. In this way, if the Bookie is wrong with 'odd', he would lose all of his money, but if the Bookies is right, he would not lose any money.

De Finetti's later definition of probability using the one-player forecasting game overcomes this problem of betting against experts, but not the first challenge of choosing rigid units of outcomes.

4.2 The Forecasting Game

The Forecasting game has a similar setup as the Prevision game, except that the one player in the game is the Forecaster.

Let $\Omega = \{\omega_1, ..., \omega_n, ...\}$ be a finite or countably infinite set of states, $\mathcal{X} = \{X_i : \Omega \to \mathbb{R} | i = 1, 2, ...\}$ be a countable set of bounded real-valued random variables defined on Ω.

The Forecaster gives a forecast $F(X)$ for each random variable $X \in \mathcal{X}$. In state ω, for each variable X, the Forecaster is penalized by a squared-error loss, i.e. $O_\omega(X, F(X)) = -[X(\omega) - F(X)]^2$. This scoring rule (squared-error) is known as the Brier Score.

The Forecaster's net score in state ω from forecasting finitely many variables $\mathbf{X} = \{X_1, ..., X_k\}$ by F is then $O_\omega(\mathbf{X}, F) = \sum_{X \in \mathbf{X}} O_\omega(X, F(X)) =$

$\sum_{X \in X'} -[X(\omega) - F(X)]^2$.

The Forecaster's forecasts $F(X) : X \in \mathcal{X}$ are *coherent*$_2$ iff there is no finite set of variables $X_1, \ldots, X_k$ and another forecast F' that yields a smaller net loss for the Forecaster in each state, i.e. there is no F' and $\mathbf{X} = \{X_1, ..., X_k\}$ such that $\forall \omega \in \Omega, O_\omega(\mathbf{X}, F) < O_\omega(\mathbf{X}, F')$.

De Finetti shows that the Forecasting game is an alternative way of defining probability. In particular, he shows that the following three statement are equivalent:

(1) A set of previsions $\{P(X)|X \in \mathcal{X}\}$ is *coherent*$_1$;

(2) The *same* set of forecasts $\{F(X)|F(X) = P(X), X \in \mathcal{X}\}$ are *coherent*$_2$;

(3) There exists a finitely additive probability $\mathbf{P}$ such that $E_{\mathbf{P}}[X] = F(X) = P(X)$ for all $X \in \mathcal{X}$.

De Finetti favors this definition of probability over the earlier version, which is subject to the challenge of betting against experts. However, this definition is also subject to the problem of choosing numeraire, which is discussed in detail in [Schervish et al., 2013]. More contrasts between these two definitions of de Finetti can be found in [Schervish et al., 2014].

Chapter 5

Group Decision

5.1 Arrow's Impossibility Theorem

Arrow's famous impossibility theorem shows that when voters have three or more distinct alternatives, the functor g, known as the social welfare rule, can convert the ranked preferences of individuals $\{\preceq_1, \ldots \preceq_n\}$ into a transitive and complete group preference $\preceq_G$ that satisfied the following four criteria: unrestricted domain, non-dictatorship, Pareto efficiency and independence of irrelevant alternatives.

(C0) Ordering: The group preference $\preceq_G$ resulting from the social welfare rule g must be transitive and complete.

(C1) Unrestricted domain: For any set of individual voter preferences $\{\preceq_1, \ldots \preceq_n\}$ over any menu of alternatives, the social welfare rule g should yield a unique and complete ranking $\{\preceq_G\}$ (over the same menu of alternatives).

(C1) is the second most blamed assumption of Arrow's theorem, as it requires the social welfare rule g to apply to all group decision scenarios. (C1) (together with C0) already excludes certain intuitive rules of group decision as a social welfare rule. For example, certain version of the pairwise majority rule.

Consider the following example of group decision with menu $\{A_1, A_2, A_3\}$ and individual preferences $\{\preceq_1, \preceq_2, \preceq_3\}$, where each individual has ordinal utilities.

	$\preceq_1$	$\preceq_2$	$\preceq_3$
A_1	1st	3rd	2nd
A_2	2nd	1st	3rd
A_3	3rd	2nd	1st

Given A_1 is strictly preferred over A_2 by $\preceq_1$ and $\preceq_3$, we have $A_2 \prec_G A_1$

(that is $A_2 \preceq_G A_1$ and $A_1 \not\preceq_G A_2$). Similarly, $A_1 \prec_G A_3$ and $A_3 \prec_G A_2$. Hence, $\preceq_G$ is not transitive since (for example) $A_1 \not\preceq_G A_2$ and $A_1 \preceq_G A_3 \preceq_G A_2$.

(C2) Unanimity (or the weak Pareto condition): If $\forall i, A_1 \preceq_i A_2$, then $A_1 \preceq_G A_2$, and if $\forall i, A_1 \prec_i A_2$ then $A_1 \prec_G A_2$.

Notice that (C2) is weaker than the strong Pareto condition (*): if $\forall i, A_i \preceq_i A_2$ and $\exists j, A_1 \prec_j A_2$, then $A_1 \prec_G A_2$.

(*) may appear to be attractive as it seems to give rise of minority rights. There are some group/social movements that are desired by some minority while the majority cannot care less. (*) allows the minority to claim their right to the group/society. However, (*) cannot be adopted without restrictions as it gives no demarcation to rights from unreasonable needs. Together with an Archimedian condition on $\preceq_G$, (*) leads to unwanted consequences.

For example, all except one member T of the group are indifferent from A_1 and A_2, while T strictly prefers A_2 over A_1. By (*), $A_1 \prec A_2$. Suppose the Archimedian condition holds for $\preceq_G$. (Recall from Savage's theory (among others) that the Archimedian condition is essential for a preference relation to have a real-value representation.) Then by the Archimedian condition, there must be some $\epsilon > 0$ such that $A_1 \prec_G A_2 - \epsilon$. That is, the group should/will be willing to pay some non-zero price *epsilon* for T's preference of A_2 over A_1. However, T's claim for A_2 to A_1 can be arbitrary. Say, it's certain idiosyncratic taste. Descriptively, every member of the group may not be willing to pay anything for T's idiosyncratic taste. Normatively, the group also shouldn't be required to pay for T's certain idiosyncratic taste. Therefore, imposing (*) in this case would allow T's idiosyncratic preference (or unreasonable needs) override the preference of all other members of the group.

Overall, (C2) is reasonably plausible and occasionally criticized.

(C3) Non-dictatorship: There is no voter i in a fixed set of voters $\{1, ..., n\}$ such that $g(\preceq_1, ..., \preceq_n) = \preceq_i$ for all possible $\preceq_1, ..., \preceq_{i-1}, \preceq_{i+1}, \preceq_n$ over a fixed menu of alternatives.

(C3) is widely found plausible and the least blamed condition proposed by Arrow.

(C4) Independence of Irrelevant Alternatives (IIA): Given a menu of choices A, a sub-menu $A' \subseteq A$ and a set of preferences $\{\preceq_1, ..., \preceq_n\}$ defined over A, the group preference over the restricted menu A' is the same as the group preference over the unrestricted A restricted to A', i.e. $g(\preceq_1|_{A'}, ..., \preceq_n|_{A'}) = g(\preceq_1, ..., \preceq_n)|_{A'}$

(C4) is the most blamed assumption of Arrow's result. In short, (C4) makes it the case that the social welfare function only cares what is preferred over another, but not to what degree it's preferred. Generally speaking, it precludes interpersonal utility comparison.[1]

[1]For more of this point, see [Luce and Raiffa, 1957], p. 344.

The following example gives a taste of this criticism.

Suppose we have a menu $A = \{A_1, ..., A_m\}$ and a sub-menu $A' = \{A_1, A_2\}$. Suppose there are 2n+1 agents, among which n+1 agents put A_1 as their 1st choice and A_2 as their 2nd choices, while the remaining n agents put A_1 as their least preferred choice and A_2 as their most preferred choice. This is shown in the following table.

	$\preceq_1$	...	$\preceq_{n+1}$	$\preceq_{n+2}$	...	$\preceq_{2n+1}$
A_1	1st	1st	1st	mth	mth	mth
A_2	2nd	2nd	2nd	1st	1st	1st
...	...	...	...	...	...	...
A_m	...	...	...	...	...	...

When we try to make the group choice over the menu A, intuitively, A_2 is a good candidate since everyone puts it either as the 2nd or 1st, while nearly half the group highly dislikes A_1. When we try to make the group choice over the sub-menu A', the information we have in the menu A that n agents in the group of size 2n+1 highly dislike A_1 in presence of other alternatives disappears. From $\preceq_i|_{A'}$, all we know is that one more person prefers A_1 over A_2 in the group. (C4) forces us to place A_1 and A_2 in the same order when deciding over A as we would for A'. That is, it requires us to give up the information we have from A and decide as if we don't have that additional information.

In general, (C4) denies any interpersonal comparison of utilities. In the example above, the first n+1 agents can have very little although non-zero preference for A_1 over A_2 and the remaining n agents can have very strong preference for A_2 over A_1. (C4) requires us to ignore these information. All that matters is that n+1 agents prefer A_1 over A_2 and n agents prefer A_2 over A_1. How much A_2 is preferred over A_1 by those n agents and how little A_1 is preferred over A_2 doesn't factor into the group's decision. This is highly implausible.

One possible way to get around Arrow's result is to deny (C4) and allow interpersonal utilities comparisons. Kevin Roberts studied extensively into this direct by studying the theory of interpersonal utilities comparisons, which clearly isn't a simple matter. See [Roberts, 1980a] and [Roberts, 1980b] for more details.

5.2 Nash's Bargaining Theory

[Nash, 1950a] gives a positive result for group decisions, known as Nash's bargaining theory, in the same year of Arrow's impossibility theorem [Arrow, 1950] and of his own vastly influential paper on Nash equilibrium [Nash, 1950b].

Nash's bargaining theory gives a unique solution to cooperative games, where participants can make binding agreements. Binding agreements in cooperative games allow participants to achieve results that are Pareto optimal but not achievable in non-cooperative games.

For example, in the Prisoner Dilemma, if we only allow pure strategy, (U,L) leads to the Pareto optimal outcome (2,2). However, as a non-cooperative game, this outcome is not a Nash equilibrium and cannot be achieved. If we also allow mixed strategies, $1/2(D, L) + 1/2(U, R)$ leads to the Pareto optimal outcome (3,3). Similarly, it's not a Nash equilibrium and cannot be achieved.

	L	R
U	(2,2)	(0,5)
D	(5,0)	(1,1)

Table 5.1: The Prisoner Dilemma

Nash's bargaining theory impose four assumptions that to some extent are similar to Arrow's. However, Nash restricts the domain of decision scenarios to "bargains" and hence gets around Arrow's results by violating (C1). In the following, I present a 2-player version of Nash's bargaining theory for convenience. The n-player version of Nash's theory has no essential difference from the 2-player version.

A bargain is a cooperative game with a default (walkaway) option (d_1, d_2) for each player. Each player on their own can walk away from the bargain. No matter what actions other players take, the walkaway of any player will make the default outcome (d_1, d_2) the actual outcome of the bargain.

	L	R	d_{Column}
U	(2,2)	(0,5)	(d_1, d_2)
D	(5,0)	(1,1)	(d_1, d_2)
d_{Row}	(d_1, d_2)	(d_1, d_2)	(d_1, d_2)

Table 5.2: A Bargaining Version of the Prisoner Dilemma

Nash supposes that all mixed-strategies are available in a bargain. A mixed strategy is a probability assignment over all pure strategies. For example, $1/2(D, L)+1/2(U, R)$ assigns 1/2 probability to (D, L), 1/2 probability to (U, R) and 0 to others. When two players decide to play $1/2(D, L) + 1/2(U, R)$, they will consult some random process, e.g. a toss of a fair coin, to decide which pure strategy, viz. (D, L) or (U, R), to take. There are philosophical problems of interpreting mixed strategies. But mixed strategies are sufficient for the purpose of maximizing expected utility. In this example, $1/2(D, L)+1/2(U, R)$ has expected utility 3 for each player and is better than every pure strategy.

A bargain is the convex set consists of all probability assignments of pure strategies. For example, the bargain of Prisoner as given in 5.2 is the area inside the red triangle in the following diagram including the boundaries, together with the special default point (d_1, d_2) to be specified.

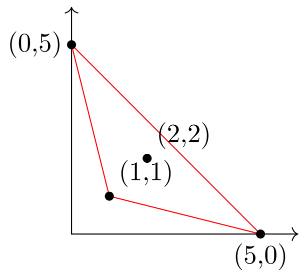


Figure 5.1: The convex set defined by Prisoner Bargain

(N1) Pareto: Let (u^*, v^*) be a solution of a bargain G, (u^*, v^*) satisfies the Pareto condition in the sense that there is no $(u', v') \in G$ s.t. $u^* < u'$ and $v^* < v'$.

As a consequence of (N1) and the existence of the default option d, the solution of a bargain is always to the northeast (upper-right) of d. (N1) also makes sure that $(1, 1)$ will never be selected over $(2, 2)$ in the bargain version of the Prisoner Dilemma. This of course is a consequence of being able to make binding agreements.

For example, suppose the default point of the Prisoner bargain is (1,1), then only strategies inside the green triangle (incl. boundaries) in the following diagram will be considered.

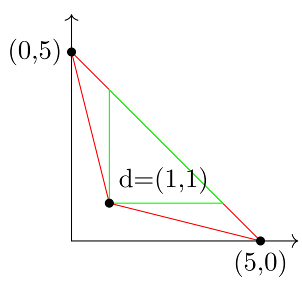


Figure 5.2: Mixed strategies allowed by (N1)

A bargain G is *symmetric* iff $d_1 = d_2$ and $\forall(a, b) \in G, (b, a) \in G$.

(N2) Symmetry: If G is symmetric, then so is the solution (u^*, v^*), i.e. $u^* = v^*$.

(N2) can be read as a condition of fairness, which is not assumed by Arrow. When each part of the bargain has the same option, the solution to the bargain should be symmetric. Formally, it's used to make sure the solution of each bargain is unique.

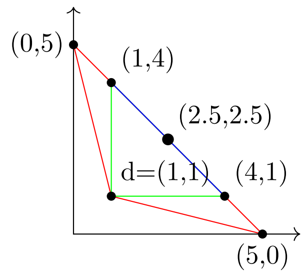


Figure 5.3: Mixed strategies allowed by (N1)

Each point on the blue line in Figure 5.3 is Pareto optimal and allowed by (N1). Only the black dot (2.5,2.5) is symmetric and hence allowed by (N2) as a solution to this bargain.

(N3) Invariance: Suppose a two-player game G is formulated with cardinal utility U, V and default choice $d = (d_1, d_2)$. The solution of G is (u^*, v^*). Take $U' = aU + b(a > 0)$ and $V' = cV + e(c > 0)$. When we formulate G using U', V' and $d' = (ad_1{+}b, cd_2{+}e)$, the solution of G would be $(u'^*, v'^*) = (au^*{+}b, cv^*{+}e)$.

(N3) serves a similar purpose as Arrow's (C4) and is hence controversial for the same reasons, i.e.it preclude interpersonal utility comparison. It doesn't matter how much one option is preferred over another for a player, as you can impose arbitrary positive linear transformation on players' utility function without changing the solution of the bargain.

Suppose Player 1 and Player 2 are trying to make a deal. Player 1 will get \$1,000,000 out of the deal while Player 2 will only get \$100. That is, consider a case where some solution is vastly preferred over others by one player but only slightly preferred by the other player. In such situation, Player 2 will intuitively have more bargaining power, since Player 1 in this case really wants to cut the deal while Player 2 cares far less. According to (N3), this bargain is no different from a bargain where Player 1 and Player 2 will both get \$1 out of the deal and nothing if one party walkaway. It's highly unintuitive that these two bargains will yield the same solution (up to the corresponding linear transformation).

(N4) Sub-game[2]: Let G^2 be a sub-game of G^1, i.e. $G^2 \subset G^1$ and G^1, G^2 have the same default option d. If (u^*, v^*) is the solution of G^2 and $(u^*, v^*) \in G^1$, then (u^*, v^*) is the solution to G^1.

(N4) is the most controversial one within the four assumptions Nash makes. A well known objection to (N4) is as follows.

The area inside the red triangle defines bargain G^1 and the yellow trapezoid inside defines the sub-game G^2.

[2]Interestingly, Nash in his original paper names this assumption "Independence of Irrelevant Alternatives" as Arrow did for his (C4), despite (N4) and (C4) serve different purposes.

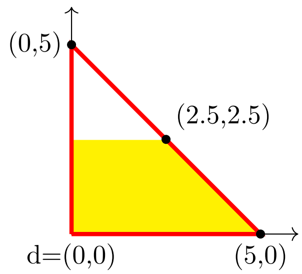


Figure 5.4: Counterexample to (N4)

One intuitive reading of this scenario is that two players are dividing a cake. If they fail to cooperate, neither of them will get any of the cake, i.e. $d = (0, 0)$. In the restricted game G^2, it's known to both players that the second player is on a diet, which makes him can take at most half of the cake.

By (N1) and (N2), the solution of the larger game G^1 is the symmetric strategy $(2.5, 2.5)$. Given (N4), the solution of G^2 must also be $(2.5, 2.5)$. However, one may find it intuitive that in G^2, player 1 has more bargaining power than in G^1. Player 1 can at least argue that in this bargain, each side should make some compromise. And if choosing $(2.5, 2.5)$, player 2 will make no comprise and get all that he ever wants. For more of this objection, see [Braithwaite, 1955], as well as [Luce and Raiffa, 1957].

Nash's positive result: Given (N1) - (N4), for every bargain G, there is a unique solution (u^*, v^*), which is the pair of u, v that maximizes the value of $(u - d_1)(v - d_2)$. This theorem works in general for n-player bargain. There, the unique solution $(u_1, ..., u_n)$ would maximize the value of $\prod_{i=1}^{n}(u_i - d_i)$.

Notice $\prod_{i=1}^{n}(u_i - d_i)$ defines a family of hyperbolic functions. From the mathematical point of view, Nash realizes that (N1) to (N4) ensures that solutions of bargains must be on hyperbolic functions. Then given every bargain as defined must be a convex set and hyperbolic functions are also convex functions, the intersect that maximizes $\prod_{i=1}^{n}(u_i - d_i)$ must be unique.

5.3 Shared Preferences of Two Bayesian Decision Makers

As mentioned earlier, Arrow's assumption (C4) Independence of Irrelevance Alternatives forces us to ignore all cardinal structures on individuals' preferences and consider ordinal utilities only. One way to recognize cardinal utilities is to consider Bayesian agents, who have a personal probability function P and a personal utility function U. We represent each Bayesian agent by a pair (P_i, U_i).

Bayesian group decision making then requires us to find a functor sending each set of Bayesian agents $\{(P_i, U_i)|i \in I\}$ (for some index set I) into a unified agent (P_G, U_G).

In the past, John Harsanyi[3] has given results showing that Bayesian agents having the same probability (i.e. $\forall i, j \in I, P_i = P_j$) but different utilities, can behave collectively as a Bayesian agent (P_G, U_G), where $P_G = P_i \forall i \in I$ and U_G is a convex combination of utility functions of each agent after certain standardization. Jacob Marschak and Roy Radner[4] on the other hand have proposed Team Theory for aggregate Bayesian agents with the same utility function but different probability functions, in a similar manner, where P_G is a convex combination of individual probabilities and $U_G = U_i \forall i \in I$.

[Seidenfeld et al., 1989] gives an impossibility result for Bayesian group decision making, from only the Pareto condition (Arrow's C2) alone. When two Bayesian agents have different subjective probability and different utility function, it's impossible for the group cannot behave like a Bayesian (or Savagian) agent without violating the Pareto condition.

Suppose we have two Bayesian agents $D = (P_1, U_1)$ and $J = (P_2, U_2)$, where $P_1 \neq P_2$ and $U_1 \neq U_2$. Without loss of generality, suppose there is an event E such that $P_1(E) = 0.1$ and $P_2(E) = 0.3$ and there are three outcomes r_0, r, r_1, such that $U_1(r_0) = U_2(r_0) = 0$, $U_1(r_1) = U_2(r_1) = 1$ and $U_1(r) = 0.1$ while $U_2(r) = 0.4$. We want to generate (P_G, U_G) from (P_1, U_1) and (P_2, U_2).

The weak Pareto condition says that whenever every individual in the group strictly prefers some act A over another act B, the group as a whole must strictly prefer A over B. Via this condition, individuals preferences over acts put constrains on the group's probability and utility functions. What [Seidenfeld et al., 1989] did is that they realize that in this case, the weak Pareto condition will exclude all but two points J and D and hence makes it impossible for two Bayesian agents to compromise.

Define f_E in the way that $f_E(E) = r_1$ and $f_E(\neg E) = r_2$. Define two (one-parameter families of) constant acts $A_{1\epsilon} := (0.1 - \epsilon)r_1 + (0.9 + \epsilon)r_0$ and $A_{2\epsilon} := (0.3 + \epsilon)r_1 + (0.7 - \epsilon)r_0$. It's easy to compute that both agents strictly prefer $A_{2\epsilon}$ over f_E and f_E over $A_{1\epsilon}$ for any positive ϵ, i.e. $\forall \epsilon > 0$, $A_{1\epsilon} \prec_1 f_E \prec_1 A_{2\epsilon}$ and $A_{1\epsilon} \prec_2 f_E \prec_2 A_{2\epsilon}$. Then by the weak Pareto condition, $A_{1\epsilon} \prec_G f_E \prec_G A_{2\epsilon}$ for all positive ϵ. Similarly, consider the constant act r and $A_{3\epsilon} := (0.1 - \epsilon)r_1 + (0.9 + \epsilon)r_0$ and $A_{4\epsilon} := (0.4 + \epsilon)r_1 + (0.6 - \epsilon)r_0$. $\forall \epsilon > 0$, $A_{3\epsilon} \prec_1 r \prec_1 A_{4\epsilon}$ and $A_{3\epsilon} \prec_2 r \prec_4 A_{4\epsilon}$. Therefore, $\forall \epsilon > 0, A_{3\epsilon} \prec_G r \prec_G A_{4\epsilon}$. Not so surprisingly, all and only points inside the yellow box (boundaries included) in the figure below satisfy both conditions, namely $\forall \epsilon$ $A_{1\epsilon} \prec_G f_E \prec_G A_{2\epsilon}$ and $A_{3\epsilon} \prec_G r \prec_G A_{4\epsilon}$. That is, the weak Pareto condition requires P_G to be a convex combination of

[3]Harsanyi, J. (1955). Cardinal Welfare, Individualistic Ethics, and Interpersonal Comparisons of Utility. Journal of Political Economy, 63(4), 309-321. Retrieved from http://www.jstor.org/stable/1827128.

[4]Jacob Marschak and Roy Radner, *Economic Theory of Teams*, Yale University Press, New Haven and London, 1972.

P_1 and P_2, and U_G to be a convex combination of U_1 and U_2.

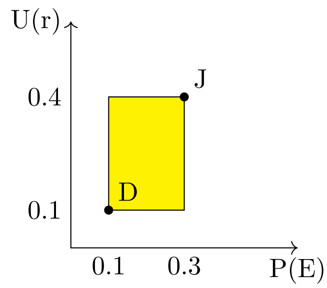


Figure 5.5: Potential compromising points of Dick and Jane.

The innovative part of [Seidenfeld et al., 1989] is that it comes up with a general way to construct a pair of (families of) acts that will rule out all points but J and D.

	E	$\neg E$
$A_{5\epsilon}$	$0.785r_0 + 0.215r_1$	$\epsilon r_0 + 0.2r + (0.8 - \epsilon)r_1$
$A_{6\epsilon}$	$(0.2 - \epsilon)r_0 + 0.8r + \epsilon r_1$	$0.165r_0 + 0.835r_1$

Table 5.3: The magical acts

Some calculation will verify that for both agents Dick and Jane, for any positive ϵ, $A_{6\epsilon}$ is strictly preferred over $A_{5\epsilon}$. For any particular ϵ, the strict preference between $A_{5\epsilon}$ and $A_{6\epsilon}$ divides the P-U plane into three parts by the rectangular hyperbola $(x - 0.2)(y - 0.25) = 0.015 - \epsilon$ with $0 < \epsilon < 0.015$. For any ϵ, the area inside two pieces of this hyperbola is where $A_{6\epsilon} \prec A_{5\epsilon}$ and the area outside is where $A_{5\epsilon} \prec A_{6\epsilon}$.

For the weak Pareto condition to be satisfied, $(P_G(E), U_G(U))$ must be inside the yellow box (including boundary) in the figure above and must be strictly outside (i.e. not on the boundary) the area surrounded by the hyperbola (that is, be in the area where $A_{5\epsilon} \prec A_{6\epsilon}$). Take $\lim \epsilon \to 0$, the area allowed by the weak Pareto condition shrinks to J and D only. For any point inside the yellow box other than J and D, there is a small enough ϵ that will rule that point out for not respecting the strict preference of $A_{6\epsilon}$ over $A_{5\epsilon}$. The strict preference for A_6 over A_5 only holds for Dick and Jane when $\epsilon > 0$. When $\epsilon = 0$, this strictly preference doesn't hold anymore. Since the weak Pareto condition only requires the group to respect unanimous strict preferences, it will never rule out J and D.

In general, for any $P_1(E) = p_1 < P_2(E) = p_2$ and $U_1(r) = u_1 < U_2(r) = u_2$, we define the pair of (families of) acts that "separates" (P_1, U_1) and (P_2, U_2) by the hyperbola $[P(E) - (p_1 + p_2)/2][U(E) - (u_1 + u_2)/2] = (p_2 - p_1)(u_2 - u_1)/4 - \epsilon$.

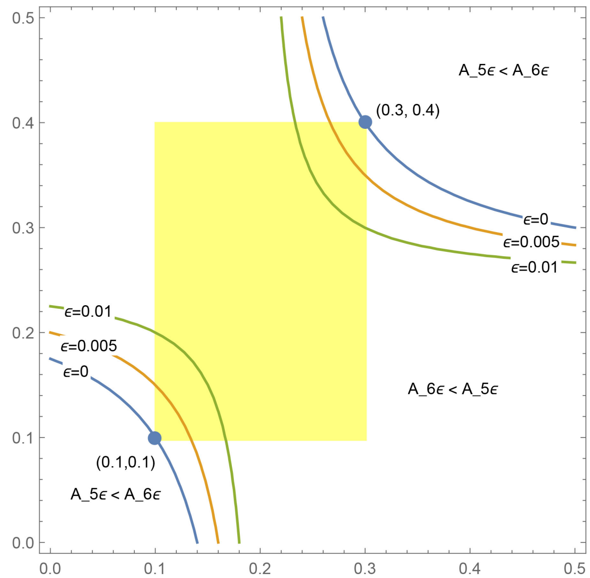


Figure 5.6: Two sets of unanimous strict preferences

From these two sets of strict unanimous preference, the weak Pareto condition excludes all points in the P-U plane except J and D as $(P_G(E), U_G(E))$. This result shows that for any two Bayesian agents with different probability and different utility, if they were to form a Bayesian/Savagian agent as a group, they can never compromise.

If we in addition assume the strong Pareto condition (which is discussed in 5.1), that if everyone either strictly prefers A over B or is indifferent between A and B, then the group strictly prefers A over B, then there will be no acceptable solution for Bayesian group decision making.[5]

This result is devastating for Bayesian group decision making. The most immediate reaction to this result is to deny the weak Pareto condition to be over strong. One may advocate some version of ϵ-Pareto, where although every individual in the group prefers B over A, the difference between A and B can be too small that no one would complain if the group chooses A instead of B. To get around this result in this ϵ-Pareto way, we, apparently, have to define what 'small' is, which isn't always easy.

[5] See [Seidenfeld et al., 1989], p. 235.

5.4 DeGroot: Reaching a consensus by sharing opinions

[Degroot, 1974] gives a non-Bayesian model for consensus. DeGroot starts with a group of agents with different subjective probabilities. Each agent in the group shares their opinions (i.e. their subjective probability) to the group and after each round of sharing, each agent will react to others' opinion by assigning probabilities to all opinions in the group (i.e. different probabilities held by members in the group).

For example, three agents start with different subjective probabilities P_1, P_2, P_3. After each round of sharing, they will replace their current probability by a weighted average of P_1, P_2, P_3. By iterate this process, considered to be a Markov process, agents will eventually reach a consensus.

Note this consensus procedure doesn't involve any Bayesian update. The consensus the group eventually gets must be inside the convex set generated by agents' the initial commitments.

Keith Lehrer and Carl Wagner wrote an entire book on such (so to speak) convex process of reaching consensus, tiled *Rational Consensus in Science and Society*. However, one may question the very idea of convex process of reaching consensus. Should the agents stay in the convex set generated by their original positions when they share opinions?

Robert Aumann gives a process of reaching consensus outside the convex set of original opinions by sharing posteriors without sharing evidence, by assuming that agents start with the same likelihood functions. (Of course, there is no magic..)

5.5 Aumann

[Aumann, 1976] discovered that a group of individuals starting with common priors can reach consensus by sharing posteriors. In the article, Aumann argues that this result indicates that people with common priors cannot agree to disagree. They will eventually reach a consensus by sharing their posteriors.

The procedure Aumann considered goes as follows:

Each individual in the group starts with a common prior and runs their own experiments. After learning from her own experiment, she announces her posterior to the group without announcing the information she gets from the experiment. This allows other individuals in the group to reverse-engineer her likelihood without knowing what she learns from the experiment.

Iterating this process, everyone's posterior will eventually become the same.

The crucial point in Aumann's procedure is that each individual is only announcing their posterior but not their information. Despite that everyone can reverse-engineer others' likelihood from others' priors and posteriors, this setting makes a difference.

	L	R
U	a_1	a_2
D	a_3	a_4

Suppose there are four states a_1, a_2, a_3, a_4. They form events U, D, L, R in the way shown in above table. Experiments run by agent B_1 allow her to learn row events (U, D) while and experiments run by agent B_2 allows her to learn column events (L, R).

The hypothesis of interest is $H = \{a_1, a_3\}$. Suppose agents B_1 and B_2 have the common prior $P_1(H) = P_2(H) = 1/2$, where P_1 is the subjective probability of B_1 and P_2 is the subjective probability of B_2. I_1 is the information B_1 gets from her experiments and I_2 is the information B_2 gets from her experiments.

After learning from her experiments, B_2 becomes certain about H either positively or negatively. That is $P_2(H|I_2)$ is either 0 or 1. On the other hand, B_1's experiment doesn't help with learning H at all, i.e. $P_1(H|I_1) = 1/2$.

After B_1 and B_2 run their experiments and announce their posterior, B_1 will learn from B_2's posterior and make her posterior equal to B_2's. A consensus is trivially achieved.

The interesting point of this example is that in this case, B_1 ends up knowing more than B_2 does. Since B_2 cannot reverse engineer B_1's information I_1, B_2 in the end only knows I_2. On the other hand, B_1 learns I_1 from experiments and learns I_2 from B_2's posterior. In the ends, B_1 gets to know exactly which state of a_1, a_2, a_3, a_4 obtains. Although ending up having different information, B_1 and B_2 nonetheless have a common posterior in the hypothesis of interest H.

Now suppose the hypothesis of interest is $H = \{a_1, a_4\}$ and B_1, B_2 have priori $P_1(H) = P_2(H) = 1/2$. Clearly neither I_1 nor I_2 helps learning about H. $P_1(H|I_1) = P_2(H|I_2) = 1/2$. B_1 and B_2 will end up both having subjective probability $1/2$ for H. However, if they share their information (instead of their posteriors), they will know that exactly which state obtains. [MORE DETAILS ABOUT AUMANN SHOULD BE ADDED.]

5.6 Bayesian Pooling

For a group of at least two Bayesian agents, the process of generating a group subjective probability P_G from individuals probabilities P_k's is called *Bayesian pooling.*[6]

[6] [Genest et al., 1986] gives a general characterization for Bayesian pooling.

For a group of size k, *linear pooling* takes a certain weighted average of all individuals' probabilities as the group's probability, i.e. $P_G = \sum_{i=1}^{k} d_i P_i$.

It turns out that linear pooling doesn't commute with conditionalization.[7] For evidence E, either every individual of the group can first update on E and then using some linear pooling to get the group probability; or we can use the same linear pooling to get the group probability from unconditionalized individual probabilities and then conditional the group probability on E. These two approaches do not always give us the same group probability conditionalized on E.

$$\begin{tikzcd}
\{P_1(\cdot), ...P_k(\cdot)\} \arrow[r, "E"] \arrow[d, "L"] & \{P_1(\cdot|E), ...P_k(\cdot|E)\} \arrow[d, "L"] \\
P_G(\cdot) \arrow[r, "E"] & P_D(\cdot|E)
\end{tikzcd}$$

Figure 5.7: Two paths don't always give the same result.

This can be illustrated with four mutually exclusive and jointly exhaustive states w_1, w_2, w_3, w_4, which form three events A, B, C and their complements in the following way.

	B	B^c	C	C^c
A	w_1	w_2	w_2	w_1
A^c	w_3	w_4	w_3	w_4

We have two Bayesian agents with probabilities P_1 and P_2. A linear pooling gives the group probability $P_G = \alpha P_1 \oplus (1-\alpha)P_2$, for $0 < \alpha < 1$.

Suppose both agents think A and B are independent, i.e. $P_1(AB) = P_1(A)P_1(B)$ and $P_2(AB) = P_2(A)P_2(B)$. It turns out the group probability P_G may not judge A and B to be independent.

All possible probabilities one can have towards w_1, w_2, w_3, w_4 are exactly the convex set generated by them, namely the regular 3-simplex (or regular tetrahedron). The independence between A and B determines a hyperbolic paraboloid (a saddle surface) inside the simplex, partitioning the simplex into two parts: one where $P(A|B) > P(A)$ and one where $P(A|B) < P(A)$.

Given both P_1 and P_2 satisfy this independence, P_1, P_2 must be on this hyperbolic paraboloid. However, a linear pooling, which corresponds to choosing a point betweeen $P_1 and P_2$ on the line determined by P_1, P_2, can make the group probability end up being outside the hyperbolic paraboloid. That is, despite both P_1 and P_2 judge A and B to be independent, P_G resulted from a linear pooling can fail to judge so. In general, $P_G(AB) = P_G(A)P_G(B)$ if and only if $P_1(A) = P_2(A)$ or $P_1(B) = P_2(B)$.

[7] Log-linear pooling doesn't have this problem. See [Genest et al., 1986].

This explains why conditionalization doesn't commute with linear pooling. If we take the blue path where we first update individual probabilities and then do the linear pooling, we will be updating on independence. One the other hand, if we first do the linear pooling, and then update, we will update on dependence.

Chapter 6

Departures from Savage's Theory

The failures of the naive Bayesian theory (i.e. a literal reading of Savage) we have seen in group decision and in randomization motivate us to take a departure from Savage's theory by weakening some of the assumptions. P1 and P2 are widely taken to be the weight-lifting assumptions of Savage and hence are the most obvious and most popular choices for rejection. Other assumptions of Savage are ancillary in nature. Denying P6 allows preferences that are cannot be represented by real numbers, e.g. lexicographic considerations as shown in section 1.3. However, it doesn't get us too far from Savage's theory. In this section, we consider and compare different ways of departing from Savage, with the focus on P1 and P2.

Denying P1 typically leads us to imprecise probability, in particular Levi's theory discussed in Section 6.3.5; while denying P2 typically leads us to prospect theory discussed in Section 6.4.

6.1 Savage on The Minimax Principle

Savage discussed two versions of the minimax principle, one violating P1 and another violating P2, with a preference for violating P1 over violating P2. Those two versions are known as the negative income form of minimax principle and the regret form of minimax principle. Savage was the first one to draw explicit distinctions between these two forms of minimax principles. For convenience and reasons we will see, we call them 'the maximin principle' and 'the minimax regret principle respectively instead.

The negative income form of minimax principle (the maximin principle) says that we should choose the act that minimizes the absolute value of the worse negative income (that is the act that maximizes the worse outcome). Given a decision matrix as follows, we should choose the row (i.e. the act) whose minimal entry is the largest among all rows. This principle comes from the study of loss functions in [Wald, 1950] and was called "minimax" since for Wald all outcomes are negative and hence can be considered as losses with positive value. This principle in that context says that one should minimize maximal loss. For reasons we will see, I will call this principle maximin instead.

	s_1	...	s_j	...	s_n
A_1	u_{11}	...	u_{1j}	...	u_{1n}
...	...	...	...	...	...
A_i	u_{i1}	...	u_{ij}	...	u_{in}
...	...	...	...	...	...
A_m	u_{m1}	...	u_{mj}	...	u_{mn}

Table 6.1: The original decision matrix

The regret form of the minimax principle (the minimax regret principle) is a variance of the maximin principle, requiring us to choose the action that minimizes the maximal difference between the outcome of chosen and the best outcome of each state.[1] Savages argues that this principle should be called "minimax loss" instead of the maximin principle, which was called "minimax loss" at that time. Savage argues that merely negative outcomes should not be called losses and instead, loss is what we now call "regret". The regret of an act in a state is the absolute value of the difference between the outcome of that act in that state and the best outcome possible in that state.

	s_1	...	s_j	...	s_n
A_1	$u_{11} - max\{u_{i1}\}$	...	$u_{1j} - max\{u_{ij}\}$	...	$u_{1n} - max\{u_{in}\}$
...	...	...	...	...	...
A_i	$u_{i1} - max\{u_{i1}\}$	...	$u_{ij} - max\{u_{ij}\}$	...	$u_{in} - max\{u_{in}\}$
...	...	...	...	...	...
A_m	$u_{m1} - max\{u_{i1}\}$	...	$u_{mj} - max\{u_{ij}\}$	...	$u_{mn} - max\{u_{in}\}$

Table 6.2: The regret matrix

Formally, the minimax regret principle requires us to first convert the decision matrix into the regret form, by shifting utilities in each column of the original matrix downward by the utility of the best outcome in the column. The regret matrix reports the difference in utility between your choice and the best choice given a state. Hence, every entry in the regret matrix is either zero or negative. In addition, each column in the regret matrix must have at least

[1] Interestingly, on page 170 of [Savage, 1954], Savage insists that the idea of minimax regret is actually from Wald despite everyone else doesn't think so. It's true that all problems Wald considered in his book are of regret form, but Savage is the first one to give a rationale for this choice of Wald.

one entry of value 0, which is the best outcome in that column (state) and admits zero regret. Then applying the maximin principle to the regret matrix will give us the act that minimize maximal regret.

In defense of this transformation into regret form, which may look unintuitive to many, Savage shows that the state-by-state shift of utility function doesn't change the preference order over acts. Take the original matrix to be defining a utility function U and the regret matrix U'. Savage shows that $E_pU'(A_i) \leq E_pU'(A_j)$ if and only if $E_pU(A_i) \leq E_pU(A_j)$.

Despite that Savage advocates minimax regret against maximin, both principles violate Savage's theory. In particular, the minimax regret principle violates P1 while the maximin principle violates P2.

Consider the following decision matrix.

	s_1	s_2
d_1	-20	0
d_2	-3	-3
d_3	-2	-5
d_4	0	-20

Table 6.3: Decision Matrix 1

It's easy to see that this matrix is in regret form. For this problem, both maximin (applying maximin to this matrix directly), and minimax regret (applying maximin to the regret form of the matrix, which is the matrix itself) give us the same result, that with all mixed strategies available, we should take the pure strategy d_2, which maximizes the worse outcome as well as the maximal regret.

Then consider the matrix with d_1 removed.

	s_1	s_2
d_2	-3	-3
d_3	-2	-5
d_4	0	-20

Table 6.4: Decision Matrix 2

Clearly, the unique maximin strategy is still d_2. However, this matrix is no long of regret form and has to be converted into the following regret form for the purpose of minimax regret.

After the transformation, the unique maximin becomes d_3. That is the minimax regret principle recommends us to choose d_3 for the decision problem represented by Decision Matrix 2, while the maximin principle recommends d_2 for Decision Matrix 2. (That is, both principles give the same answer for Decision Matrix 3, namely d_3, but different answers for Decision Matrix 2.)

	s_1	s_2
d_2	-3	0
d_3	-2	-2
d_4	0	-17

Table 6.5: Decision Matrix 3

In this example, the minimax regret principle violates property α, that whatever not chosen in the original menu is not chosen in the extended menu. Here, take the original menu to be the acts available in Decision Matrix 2, namely $\{d_2, d_3, d_4\}$ and the extended menu to be acts available in Decision Matrix 1, namely $\{d_1, d_2, d_3, d_4\}$. d_2 is not chosen in the original menu but chosen in the extended menu. Given α is necessary for P1, the minimax regret principle violates P1.

The violation of P2 by the maximin principle is more straightforward. Consider the following matrix.

	s_1	s_2
A	0	1
B	1	0

By the maximin principle, A and B are indifferent. By P2, $0.5A + 0.5B \approx 0.5B + 0.5B = B$. However, the expected minimal outcome of $0.5A + 0.5B$ is $1/2$ greater than that of B. So by the maximin principle $0.5A + 0.5B \succ B$. Hence, P2 is violated.

Savage gives an argument for minimax regret over maximin in pp. 170-171, using the following example.

	s_1	s_2
f_1	-1	-1
f_2	-10	1

For this matrix, the pure strategy f_1 is recommended by the maximin principle. If one follows the maximin principle, unless being certain, no matter what one learns about the probability of s_1, one will always choose f_1 for maximizing worse outcome. The absurdity of this feature of maximin is emphasis by the following example of Savage.

> A person has a ladder, and, just as he is about to use it, it occurs to him that the ladder may possibly be dangerously defective. He envisages two basic primary acts: f_1 , to throw the ladder away and buy a new one, which will cost 1 utile in either event; and f_2, to use the ladder, which will, if the ladder is defective, result in his injury to the extent of 10 utiles, and will, if the ladder is sound, accomplish

> his object, which is worth 1 utile. Now, if the person acts on the principle of minimizing the maximum of negative income, he will throw the ladder away, no matter what tests tend to show that it is sound.

This feature of the maximin principle is called ultrapessimistic for leading to the ignoring of any non-certain evidence. An alternative way to see its absurdity is that all non-certain evidence about s_1 is judged to be worthless in this problem for it will not influence your decision at all, which is apparently counterintuitive. On the other hand, if we transform this matrix into regret form, the maximin strategy in the regret matrix would be a non-pure mixed strategy of f_1 and f_2. In that case, our evidence in s_1 can guide us on which mixed strategy to choose.

However, this argument for the minimax regret principle doesn't work. It turns out that this problem is not unique to the negative income form of the minimax principle, but also to the regret form. This problem is explored in detail in [Parmigiani, 1992]. I present the example given in that paper, where the minimax regret principle is ultrapessimistic while the maximin principle is not.

	RR	BR	BB
a_1	-2	0	-4
a_2	-4	-4	0

The outcome in the decision matrix is decided by a sequential observation of B and R. Only three configurations, namely RR, BR and BB, are possible. It can be calculated that the unique maximin strategy is $2/3a_1 + 1/3a_2$, with $-(2 + 2/3)$ for RR and BB, $-(1 + 1/3)$ for BR.

One is allowed to take the experiment to learn the first observation at the cost 0.1. Then one has to make a sequential decision, first whether to take the experiment, and second what strategy to make for the decision. If one learns R from the experiment, then one would choose a_1 for sure for being certain about RR. If one learns B from the experiment, then one may choose a_1 or a_2, for a lack of dominance. This sequential decision can be represented as the following table, where a_3 and a_4 are the acts of taking a_1 and a_2 respectively after observing B from the experiment. Notice $a_3(RR) = a_4(RR) = -2.1$. It's because, if RR is the case, then one would learn R and choose a_1 for sure. The difference between a_3 and a_4 is what one chooses after learning B.

	RR	BR	BB
a_1	-2	0	-4
a_2	-4	-4	0
a_3	-2.1	-0.1	-4.1
a_4	-2.1	-4.1	-0.1

Some computation will show that the unique maximin strategy is $0.5a_1 +$

$0.5a_2$, which recommends one to take the experiment with positive probability.

Now we consider the regret form of the decision matrix.

	RR	BR	BB
a_1	0	0	-4
a_2	-2	-4	0

The maximin strategy in the regret matrix is $0.5a_1 + 0.5a_2$, with -1 for RR and -2 for others. Then consider the regret matrix augmented by the choices of taking the experiment.

	RR	BR	BB
a_1	0	0	-4
a_2	-2	-4	0
a_3	-0.1	-0.1	-4.1
a_4	-0.1	-4.1	-0.1

In this matrix, the RR column doesn't make any contribution to the consideration of maximin for being dominated by BR (i.e. the regrets in BR column are always larger than those in RR and hence it's sufficient for the purpose of minimizing maximal regret to minimize regrets in BR with regrets in RR ignored). Then, without RR ignored, a_3 is dominated by a_1 and a_4 is dominated by a_2. a_3 and a_4 are not admissible in this matrix according to the Maximin principle, and hence they are not admissible according to the minimax regret principle for the original problem.

This example shows that one intuitively valuable experiment is judged to be worthless by the minimax regret principle, but by the maximin principle.

Giovanni, in his paper, goes on to characterize what kind of experiments will be judged to be worthless by each principle. It turns out that both versions of the minimax principle suffer from Savage kind ultrapessimistic arguments equally.[2]

6.2 Seidenfeld's Argument for Rejecting P1

[Seidenfeld, 1988] gives an argument for rejecting P1 instead of P2, on the basis that once P2 is rejected, in sequential decision problems, decisions are not invariant under the substitution of outcomes judged to be equivalent.

[2]One may have the concern that Savage's original argument, although sketchy, argues that no experiment will be judged valuable by the maximin principle, while Giovanni only gives conditions for some kind of experiments to be judged by worthless by each condition. In fact, Savage is also only discussing a certain kind of experiments, namely the kind that changes your probability without making you certain about the quality of the first ladder. Experiments that make you certain about the quality of the first ladder are not quantified over by Savage. Indeed, Savage's consideration and Giovanni's considerations are of the same kind.

Consider the sequential decision problem defined in the following graph. The agent judges L_1 and L_2 to be equivalent in value, but strictly prefers the half-half mix of L_1 and L_2, for maximin consideration, i.e. $L_1 \approx L_2$; $0.5L_1 + 0.5L_2 \succ L_1$. Notice this violates P2, as P2 allows us to derive that $0.5L_1 + 0.5L_2 \approx L_1$, from $L_1 \approx L_2$ by $L_1 = 0.5L_1 + 0.5L_1$.

Then we suppose lotteries under consideration, namely L_1, L_2, $0.5L_1 + 0.5L_2$ to have certain monetary value. Without loss of generality, say $L_1 \approx L_2 \approx \$5$ and $0.5L_1 + 0.5L_2 \approx \$6$. Suppose there is some small but positive transaction fee ϵ one needs to pay for running the mixed lottery $0.5L_1 + 0.5L_2$.[3] Suppose $\epsilon \approx \$0.25$. Then the mixed strategy $0.5L_1 - \epsilon + 0.5L_2 - \epsilon$ has value \$5.75.

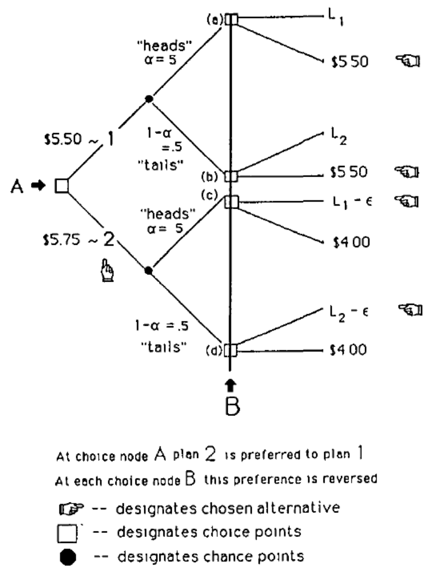


Figure 6.1: A sequential decision problem

The agent at point A chooses which branch to go (upper half or lower half). Then, after tossing a fair coin, the agent gets to choose again at B. Using backward induction, the agent knows that at (c) or (d), she will choose $L_1 - \epsilon$ and $L_2 - \epsilon$ respectively, since they both have value \$4.75 greater than \$4. Then standing at A, the agent sees the lower half (going down) as the mixed lottery $0.5L_1 - \epsilon + 0.5L_2 - \epsilon$, which makes the down branch to have value \$5.75. Similarly, using backward induction, the agent will choose \$5.5 at both (a) and (b). Therefore, the agent sees the upper half of the tree as the constant outcome \$5.5. The agent will then choose to go down for higher expected value.[4]

[3] As it seems to me, this transaction fee is mainly for realistic purposes, and isn't essential for the argument to go through.

[4] The backward induction strategy used here is sometimes called "sophisticated choice". One may find it strange why the agent doesn't take the upper half as the mixed lottery

Suppose we replace $L_1 - \epsilon$ and $L_2 - \epsilon$ in the game by their monetary value $\$L_1 - \epsilon \approx \$L_2 - \epsilon \approx \$4.75$, as shown in the following figure. Then the agent will choose the upper half instead of the lower half, at A. This example, then, shows that the sequential decision isn't invariant under the substitution of outcomes judged to be equivalent by the ordering.

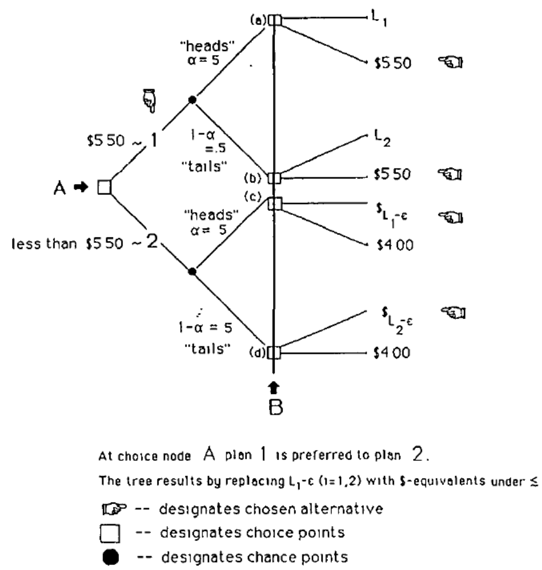


Figure 6.2: The same problem with equivalent outcomes substituted

Seidenfeld argues that this puts people who reject P2 (independence) in order to keep P1 (ordering) in an awkward situation where the ordering isn't really respected. Namely, outcomes judged to be equivalent by the ordering (in the sense that neither is preferred over the other) fail to be inter-substitutable .

People who deny P2 typically have to reply to this argument by emphasizing that it's the very purpose of denying P2 to deny the inter-substitutability of outcomes judged to be of equal value.

One particular reply in this spirit is given by Mark Mashina. Mashina argues that the choice of $L_1 - \epsilon$ at (c) is non-separable from the choice of $L_2 - \epsilon$ at (d). Intuitively, the agent chooses to go the lower branch not because of $L_1 - \epsilon$ or $L_2 - \epsilon$ individually, but because of the combining effect of them. What the agent is choosing at (c) is not the outcome $L_1 - \epsilon$ simpliciter, but is rather $L_1 - \epsilon$ under the context of the lower branch, more specifically, under the context of

---

$L_1 + L_2$, which has value \$6. Doing so requires the agent to choose L_1 or L_2 over \$5.5 at (a) or (b) respectively despite \$5.5 is preferred over L_1. This sequential decision strategy is called "resolute choice" and is defended extensively in [McClennen, 1990]. However, whether one is making sophisticated choices or resolute choices doesn't make a difference to the argument. For a resolute choice maker, the same point can be made by substituting L_1 and L_2 with their monetary value \$5.

the presence of $L_2 - \epsilon$ in a certain manner. Therefore, in this case, $L_1 - \epsilon$ under such context isn't equivalent to and hence cannot be substituted by \$4.75, even though the outcome $L_1 - \epsilon$ on its own is indeed equivalent to \$4.75.

6.3 Imprecise Probabilities

One departure from Savage is theories of imprecise probabilities, where the agent is allowed to have a set of probability functions to reflect their uncertainties, instead of a single probability function.

Imprecise probability alone doesn't guide one on how to make decisions. It only becomes a decision theory after accompanied with certain decision rules. After discussing how to elicit imprecise probabilities, I will discuss three different rules one may adopt to guide decisions in theories of imprecise probabilities.

6.3.1 De Finetti's Fundamental Theorem of Prevision

An early instance of imprecise probability can be found from de Finetti's fundamental theorem of prevision. This theorem de Finetti is an extension theorem for previsions in his prevision game (See Section4.1 for the prevision game).

In the original prevision game, the bookie gives a prevision function (which is probability function if it's coherent) defined for each variable in a set, both for selling and buying. Such previsions, used for both selling and buying, are called 2-sided previsions (in contrast to 1-sided previsions to be introduced in later sections). De Finetti shows that any coherent 2-sided prevision can be extended without failing to be coherent. [5]

Suppose the bookie has a coherent prevision function over a set of variables χ. This forces the agent to have previsions on all linear combinations of χ, which is the span of χ, noted by $Span\chi$.

Take variable $Y \notin Span\chi$. We want to extend the existing previsions over χ to coherent previsions over $\chi \cup \{Y\}$.

We can approximate Y using elements in $Span\chi$ from below and above by

$$\underline{A} := \{X \in Span\chi | (\forall\omega) X(\omega) \leq Y(\omega)\}$$

and

$$\bar{A} := \{X \in Span\chi | (\forall\omega) X(\omega) \geq Y(\omega)\}.$$

Let $\underline{P}(Y) = \sup_{X \in \underline{A}} P(X)$ and $\bar{P}(Y) = \inf_{X \in \bar{A}} P(X)$. For the resulting The 2-sided prevision after extension to be conherent, $P(Y)$ must be in the interval $[\underline{P}(Y), \bar{P}(Y)]$.

[5] More details can be found in [De Finetti and de Finetti, 1974].

This result of de Finetti can be seen as a very early version of imprecise probability. However, it's too weak to serve most purposes one may want to achieve using imprecise probability. For example, it doesn't allow us to make the Ellsberg choice (See 1.4.2), since as soon as the agent chooses Gamble 1 over Gamble 2, this choice reveals that the agent's probability on Blue is higher than on White and forces the agent to choose Gamble 3 over Gamble 4.

This result overall can only be seen as an incomplete elicitation of precise probabilities, but not an account of imprecise probability. One may interpret the result as follows: agent A has the precise probability over $\chi \cup \{Y\}$. I, however, only observed her previsions over χ but not Y. Then by this result, I can know that her prevision on Y must fall into a certain interval. This doesn't change the fact that she has a precise prevision on Y. But this is a really weak interpretation. A stronger (i.e. leaning more towards the direction of imprecise probability) interpretation is that the agent so far only has precise previsions on χ. To be rational, she must also have a precise prevision on Y once Y joins the trade. This result shows that such extension is possible, without telling you how the agent gets that precise prevision on Y.

6.3.2 Eliciting Imprecise Probabilities: the Generalized Prevision Game

In the original prevision game, for each random variable, the Bookie announces a prevision (fair price) for buying and selling. Imprecise probabilities can be elicited using a generalized version of the prevision game, where the bookies is allowed to announce two price, one for sell, one for purchase. Such previsions for sell/purchase only but not both are known as 1-sided previsions.

[Smith, 1961] proposes to generalize the prevision game so that for each random variable x the bookie is allowed to announce two prices, $\bar{P}(x)$ for sell and $\underline{P}(x)$ for purchase. Smith shows that the customer cannot make a Dutch book against the bookie if and only if there is a set of probabilities $\mathcal{P}$, s.t. for each random variable x, the bookie's prevision for sell $\bar{P}(x)$ equals the supreme of the set of probabilities, i.e. $\bar{P}(x) = \sup(\mathcal{P}(x))$, and the bookie's prevision for purchase $\underline{P}(x)$ equals the infimum of the set of probabilities, i.e. $\underline{P}(x) = \inf(\mathcal{P}(x))$.

This is a generalization of de Finetti's result. When the bookie sets the two previsions to be the same, $\mathcal{P}$ becomes a singleton set; and when the two previsions are different, this game elicits a set of probabilities instead of a single probability.[6]

Once the agent is allowed to have imprecise probabilities, the agent's uncertainty over events becomes incomparable in some cases. For example, suppose $\mathcal{P}(A) = [0.3, 0.7]$ and $\mathcal{P}(B) = [0.5, 0.8]$. Although it seems that $\mathcal{P}$ judges B to

[6] R. Nau (1985) further generalizes this game to allow the bookie has different previsions depending on the size of the trade.

be more probable than A, the most natural/secured/popular interpretation is still that one's uncertainties in A and B are not comparable.

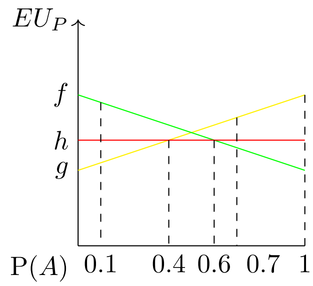
6.3.3 Γ-maximin

The decision rule for imprecise probabilities proposed by C.A.B. Smith is the Γ-maximin rule, which recommends one to choose the act that maximizes the minimal outcome over the set of probabilities $\mathcal{P}$. That is for a set of acts S, $\mathcal{C}(S) = \{f \in S | min_{P \in \mathcal{P}}(P(f) * U(f)) = max_{f \in S}(min_{P \in \mathcal{P}}(P(f) * U(f))\}$.

Consider three acts f, g, h as in the following chart.

	A^C	A
f	1	0
g	0	1
h	0.4	0.4

Suppose the agent's uncertainty on A, $\mathcal{P}(A) = [0.1, 0.7]$. The expected value of each act is illustrated in the following figure.



Given $\mathcal{P}(A) = [0.1, 0.7]$, g reaches its minimal expected value when $P(A) = 0.1$, which is 0.1 and f reaches its minimal expected value when $P(A) = 0.3$, which is 0.3. That is, $min_{P \in \mathcal{P}}(E_P(f)) = 0.3$ and $min_{P \in \mathcal{P}}(E_P(g)) = 0.1$. For any $P \in \mathcal{P}$, $E_P(h) = 0.4$. According to Γ-maximin, one should have the preference $g \prec f \prec h$.

Smith's theory is one step further towards imprecise probability from de Finetti. For example, it gives us the Ellsberg choice. It can be easily verified that according to Γ-maximin, Gamble 1 is chosen over Gamble 2 while Gamble 4 is chosen over Gamble 3.

It's worth-noting that Γ-maximin alone as a decision rule gives a complete weak order over acts, despite that imprecise probabilities make agent's uncertainties fail to be comparable in general. This is not hard to see since for each act, Γ-maximin only cares about its minimal expected value in respect to a given set of probabilities. As it turns out, imprecise probability supplemented with Γ-maximin alone gives a decision theory that satisfies P1 but violates P2.

The violation of P2 can be easily seen from the above example. $0.5g \oplus 0.5f$ has value 0.5 greater than $0.5f \oplus 0.5f$, despite that $g \prec f$.

An Argument from De Finetti and Savage

De Finetti and Savage have a co-authored paper on imprecise probability written in Italian ([De Finetti and Savage, 1962]), where they argue against C.A.B. Smith's account. The content of that paper and further discussions can be founded in [Vicig and Seidenfeld, 2012].

Their argument roughly goes as follows: Suppose, for event E, the bookie has 0.7 as selling price and 0.4 as buying price and hence, for $\neg E$ has 0.6 as selling price and 0.3 as buying price. Suppose one customer wants to buy E at 0.6 while another will buy $\neg E$ at 0.5. According to C.A.B. Smith, the bookie will not make deals any of these two customers, since bookie only makes deal at the announced price. However, these two deals together form an arbitrage. The bookie will win 0.1 for sure by making these two deals together.

[Vicig and Seidenfeld, 2012] replies to this argument relying on the menu-dependence nature of imprecise probability. Whether the bookie makes or doesn't make certain deals depends on what deals are available. This nature of imprecise probability will be more clear later in discussing Levi's theory.

6.3.4 Maximality

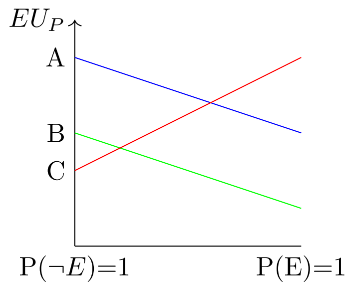
Another principle considered by [Walley, 1991] among others for decision making using imprecise probabilities is the priniciple of maximality.

Say an act A dominates act B with respect to a set of probability $\mathcal{P}$ iff for every $P \in \mathcal{P}$, $E_P(A) > E_P(B)$. The principle of maximality says that For menu M, the set of admissible choice $\mathcal{C}(M)$ is those acts in M that are not dominated by any act in M, i.e. $\mathcal{C}(M) = \{x \in M | \forall y \in M, y \text{ doesn't dominate } x\}$.

Maximality as a decision rule violates property β, but satisfies α and γ, and hence allows pairwise comparison. This should not be surprised, as this principle is essentially asking you to pairwise compare all acts and eliminate those lose at some battle.

This feature of Maximality can be brought out by the following example.

According to Maximality, $\mathcal{C}(A, B, C) = \{A, C\}$; $\mathcal{C}(A, C) = \{A, C\}$; $\mathcal{C}(B, C) = \{B, C\}$, $\mathcal{C}(A, B) = \{A\}$. This violates property β, since in the menu $\{B, C\}$, both B and C are chosen. β requires in this case that when the menu is extended, either both B and C are chosen or neither is. However, when we extend $\{B, C\}$ by A, C is chosen while B is not. This doesn't violate γ or α. For



example, consider $\mathcal{C}(A, C) = \{A, C\}$; $\mathcal{C}(B, C) = \{B, C\}$. Only A wins all (in this case, two) local tournaments and is required by γ to be a winner of the grand tournament $\{A, B, C\}$.

So the rule of maximality, although violates β, nonetheless satisfies α and γ, which makes it a 'normal' choice rule according to Sen (see 1.3.2).

6.3.5 E-admissibility

Issac Levi proposes E-admissibility as a decision rule for imprecise probability, called. Using E-admissibility alone gives us a theory where P1 is violated and P2 is (in some sense) respected.

The principle of E-admissibility saying that a choice is admissible if it's optimal for some probability in your set of probabilities.[7]

Recall that P1 requires one to have a weak order over acts, which is reflexive, transitive, complete and context-free preference over acts. Denying P1 opens the door for one's preference over acts to be incomplete. That is, it allows acts to be incomparable. Theories denying P1, hence, naturally shift their considerations to what is admissible from what is unconditionally optimal. That is, such theories often aim to give you a set of admissible choices, instead of one optimal choice.

As a consequence, E-admissibility doesn't always give you a unique choice but often recommend more than one admissible choices for you to choose from. For this reason, Levi recommends to combine some other decision rules, e.g. Γ-maximin, with E-admissibility so that one first apply the principle of E-admissibility and then use other rules to choose within the E-admissible options.

Earlier in section 1.3, we mentioned that P1 is equivalent to Sen's property α and β. Levi's E-admissibility violates property β and hence violates P1.[8] In addition, it violates γ, which is a property of choice functions weaker than β in

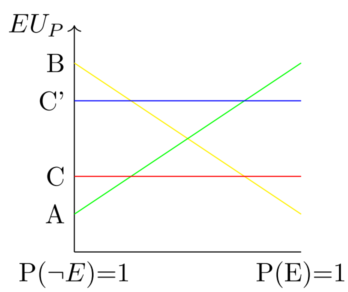
[7] This idea of E-admissibility can also be found in [Savage, 1954], pp. 122-124.

[8] Supplemented by Γ-maximin, E-admissibility also violates property α. See [Seidenfeld, 2004] for detail.

presence of α, discussed earlier in Section 1.3.2.

One of the greatest disadvantage of Levi's theory of imprecise probability is its violation of property γ. As discussed earlier (in Section 1.3.2), property γ allows us to determine the preference over a menu by pairwise (or other local) comparisons.

Consider four acts A, B, C, C' defined as in the figure below. You would never choose C when A and B are both available, since C is never optimal. For any P, either A or B is strictly better than C. On the other hand C' is in some case (i.e. for some P) optimal.



Note the choice function by $\mathcal{C}$. According to E-admissibility, $\mathcal{C}(A, B, C) = \{A, B\}$ while $\mathcal{C}(A, B, C') = \{A, B, C'\}$[9] . By E-admissibility, $\mathcal{C}(A, C) = \{A, C\}$ and $\mathcal{C}(B, C) = \{B, C\}$.

This already demonstrates the violation of β and γ from E-admissibility. Both A and C were admissible in the original menu $\{A, C\}$, while only A but not C is admissible in the augmented menu $\{A, B, C\}$. This violates property β. C is admissible from the sub-menus $\{A, C\}$ and $\{B, C\}$, but not from the grand-menu $\{A, B, C\}$. This violates property γ.

In addition to these violations, it helps to note that the difference between C and C' cannot be brought out by pairwise comparisons. By E-admissibility, $\mathcal{C}(A, C') = \{A, C'\}$ and $\mathcal{C}(B, C') = \{B, C'\}$. Despite that $\mathcal{C}(A, B, C) = \{A, B\}$ but $\mathcal{C}(A, B, C') = \{A, B, C'\}$, pairwise comparisons within $\{A, B, C\}$ and $\{A, B, C'\}$ give us the same information for C and C'.

Earlier, we mentioned that Levi's theory is an example of violating P1 without violating P2. This is subtler than it sounds. Recall that P2 involves the ordering $\preceq$ in its formulation. Whether Levi's theory respects P2 or not depends on how the $\preceq$ is defined for Levi's theory. [Seidenfeld, 1988] gives a definition of categorical preference, that $f \prec_C g$ iff whenever $f, g \in S$, $f \notin \mathcal{S}$. This means the presence of g always excludes f from being chosen. Levi's theory (with E-admissibility alone or with some particular supplementary rules) respects P2 with respect to the relation $\prec_C$ and is also immune from Seidenfeld's argument

[9]For convenience, I use $\mathcal{C}(X_1, ..., X_n)$ as an abbreviation for $\mathcal{C}(\{X_1, ..., X_n\})$.

discussed earlier in Section 6.2.

From Levi's theory, we can see how much menu dependence one will face once P1 is violated. Economists often prefer to deny P2, which typically leads to prospect theory, to avoid menu dependence and save the method of pairwise comparison.

6.4 Prospect Theory

Another departure from Savage is to reject P2 and keep P1 (and hence keep the context-free-ness of preferences). Prospect theory is one of the most popular choices. In the following, I give a cartoon version of the original prospect theory.[10].

Prospect theory is intended to be a descriptive theory explaining preferences of people that cannot be explained by expected utility theory, e.g. risk-averse behaviors, the seemingly intuitively choices in the Ellsberg paradox (see section 1.4.2 for the paradox), etc. The idea of prospect theory is that in addition to probability, we have a subjective prospect function Π over probabilities. For example, a risk-seeking agent, who prefers to take risks for better outcomes (despite the same expected utility), tend to give higher prospect to low probabilities.

The prospect function is a monotone non-decreasing function sending $[0, 1]$ to $[0, 1]$. When facing a decision problem, we compute the prospect values of acts, which are the products of prospect and utility, instead of the expected values of acts, which are the products of probability and utility. When the prospect function is the identity function from $[0, 1]$ to $[0, 1]$, the agent's prospect degenerates to the agent's probability and hence prospect theory degenerates to expected utility theory. For example, $\Pi_1(x) = x^2$ and $\Pi_2(x) = \sqrt{x}$ are both prospect functions.

Consider the following example, where a_1, a_2 are two lotteries and the prospect function Π is such that $Pi(0.5) = 0.25$ and $\Pi(0.25) = 0.1$.

	s_1	s_2	s_3
$P(s_i)$	0.5	0.25	0.25
a_1	0	-8	-8
a_2	0	-9	-10

With probability 0.5, a_1 returns outcome -8 and 0 otherwise. With probability 0.25, a_2 returns outcome -9, with probability 0.25, outcome -10, and 0 otherwise.

The prospect of a_1 is $\Pi(0.5) * (-8) = 0.25 * (-8) = -2$, while the prospect of a_2 is $\Pi(0.25)*(-9)+\Pi(0.25)*(-10) = 0.1*(-9)+0.1*(-10) = -1.9$. Therefore,

[10]For more on (cumulative) prospect theory, see [Wakker, 2010]

prospect theory recommends a_2 for having a higher prospect value. However, notice that if we consider a_1 and a_2 to be acts, then a_1 dominates a_2 state by state. If we consider a_1 and a_2 as lotteries (i.e. with utilities and probabilities specified, but not defined by states), then a_2 is stochastically dominated by a_1.[11]

Stochastic dominance is a generalization of state by state dominance: gamble A has first-order stochastic dominance over gamble B if for any outcome x, A gives at least as high a probability of receiving at least x as does B, and for some x, A gives a higher probability of receiving at least x. In notation form, $P[A \geq x] \geq P[B \geq x] P[A \geq x] \geq P[B \geq x]$ for all x, and for some x, $P[A \geq x] > P[B \geq x] P[A \geq x] > P[B \geq x]$.

It turns out that the violation of first-order stochastic dominance is characteristic of the original prospect theory. This violation motivates economists to move to cumulative prospective theory, where prospect functions are defined using the cumulative probability distribution function, as in rank-dependent expected utility theory, so that the first-order stochastic dominance is preserved.[12]

Overall, economists have been seeking clever ways to violate independence without violating stochastic dominance, since doing so (i.e. violating independence without violating stochastic dominance) doesn't force one to run into sure-loss in single-shot decisions, i.e. isn't subject to Dutch-book for non-sequential decisions.

One insight on stochastic dominance is as follows:

Consider three outcomes $r_1 \prec r_2 \prec r_3$ forming a convex set (a equilateral triangle) by convex combination, as follows.

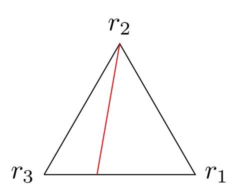


Figure 6.3: Indifference curve for r_2 in Expected Utility Theory

The red line is an indifference curve (containing r_2), where all points on the curve are of the same value. For expected utility theory, all such indifference curves are straight lines. Departing from expected utility theory, prospect

[11]It's easy to see how prospect theory violates P2 from a slight variance of this example. Replace -10 in the example by -8. Then from P2, it follows that $a_2 = 0.5 * 0 + 0.25 * 0.8 + 0.25 * 0.9 \prec a_1 = 0.5 * 0 + 0.25 * 0.8 + 0.25 * 0.8$. However, it can be easily computed that a_2 has a higher prospect than a_1 and is hence recommended by prospect theory over a_1.

[12]The first correct fix of the original prospect theory is due to John Quiggin. See his book *Generalized Expected Utility Theory.* The original fix given by Tversky and Kahneman (1992) turns out to be flawed.

theory allows us to have indifference curves that are not straight lines.

In order for prospect theory to respect stochastic dominance, the indifference line has to satisfy some restrictions.

Take an arbitrary point in the convex set, as in the following graph.

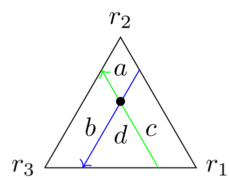


Figure 6.4: Stochastic Dominance in the Convex Set

Along the green line, in the direction of the arrow, the proportion of r_3 doesn't change while the proportion of r_2 increases and the proportion of r_1 decreases. Therefore, on the green line, every point stochastically dominances any point to its right. Similarly, along the blue line, every point stochastically dominances any point to its right.

These two lines of stochastic dominance partitions the triangle into four areas, a, b, c, d. Every point in b stochastically dominates the middle point, while every point in c is stochastically dominated by the middle point. For a theory to respect stochastic dominance, the indifference line of the middle point must be in area a and area d.

Chapter 7

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